

# Predicting Student Dropout Risk

An explainable and fair early-warning model for targeted student support

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Integrated report covering capstone Steps 1–5: problem framing, data understanding, exploratory analysis and feature engineering, model implementation and comparison, and explainability, bias and fairness analysis.

Dataset: Predict Students' Dropout and Academic Success (UCI Machine Learning Repository, dataset 697; 4,424 students).

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**Code repository:** <https://github.com/Bunz825/student-dropout-prediction>

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# Contents

|                                                             |           |
|-------------------------------------------------------------|-----------|
| <b>Executive summary</b>                                    | <b>3</b>  |
| <b>About this report</b>                                    | <b>4</b>  |
| <b>1. Problem Understanding and Framing</b>                 | <b>5</b>  |
| 1.1 Business context                                        | 5         |
| 1.2 Data science framing                                    | 6         |
| 1.3 Task type and justification                             | 6         |
| 1.4 Success metrics (model)                                 | 6         |
| 1.5 Business KPIs                                           | 7         |
| 1.6 Capstone linkage                                        | 8         |
| <b>2. Dataset Overview and Data Dictionary</b>              | <b>10</b> |
| 2.1 Data source and collection                              | 10        |
| 2.2 Dataset overview                                        | 11        |
| 2.3 Data quality: missing values, duplicates and outliers   | 12        |
| 2.4 Data dictionary                                         | 13        |
| 2.5 Limitations and how later steps address them            | 14        |
| <b>3. Exploratory Data Analysis and Feature Engineering</b> | <b>15</b> |
| 3.0 Reproducible code                                       | 15        |
| 3.1 Data cleaning                                           | 15        |
| 3.2 Applied exploratory data analysis                       | 16        |
| 3.3 Feature engineering                                     | 18        |
| 3.4 Feature importance and explainability                   | 20        |
| 3.5 Feature selection                                       | 20        |
| 3.6 Dimensionality reduction                                | 21        |
| 3.7 Preprocessing pipeline and leakage controls             | 22        |
| <b>4. Model Implementation and Comparison</b>               | <b>23</b> |
| 4.1 Experimental design                                     | 23        |
| 4.2 Models implemented and tuned                            | 23        |
| 4.3 Supervised results and comparison                       | 24        |
| 4.4 Unsupervised learning                                   | 25        |
| 4.5 Recommendation systems — rationale for not using        | 26        |
| 4.6 Deep learning — scope and rationale                     | 26        |
| 4.7 Model selection and reasoning                           | 27        |
| 4.8 Reproducibility                                         | 28        |
| 4.9 Limitations                                             | 28        |
| <b>5. Explainability, Bias and Fairness Analysis</b>        | <b>29</b> |
| 5.0 Link to Chapter 4 and scope of the audit                | 29        |
| 5.1 Explaining model decisions                              | 29        |
| 5.2 Limitations: imbalance, leakage and overfitting         | 31        |
| 5.3 Bias detection and fairness audit                       | 31        |
| 5.4 Mitigation strategies tested                            | 33        |
| 5.5 Responsible deployment and monitoring                   | 35        |
| <b>6. Conclusions and Recommendations</b>                   | <b>36</b> |
| 6.1 Answering the business question                         | 36        |
| 6.2 Recommendations                                         | 36        |
| 6.3 Limitations and future work                             | 36        |
| 6.4 Closing statement                                       | 37        |
| <b>References</b>                                           | <b>38</b> |
| <b>Appendix A — Code lists for nominal variables</b>        | <b>39</b> |
| <b>Appendix B — Repository and reproduction guide</b>       | <b>43</b> |

## Executive summary

Almost one in three students (32.1%) at an educational institution leaves before completing their degree. Advisors cannot support every student intensively, and today they usually learn who is struggling only after a student has disengaged. This project builds an early-warning model that uses information available by the end of a student's first year to estimate each student's probability of dropping out, so that advisors can offer help to the right students early enough to make a difference.

The model was built on the public, peer-reviewed UCI dataset of 4,424 students (36 variables covering demographic, socioeconomic, macro-economic, enrolment and academic-performance data). The data is complete, with no missing values or duplicates. Twenty-two features were engineered and 25 were selected. Five supervised models and four unsupervised techniques were then compared on one shared, seeded data split. The selected model was explained with SHAP, LIME, PDP and ICE and audited for fairness across gender, age, scholarship, debtor and international status.

### Results against the success targets

| Success target (Chapter 1)                      | Result                                                                     | Met?          |
|-------------------------------------------------|----------------------------------------------------------------------------|---------------|
| Recall (dropouts caught) $\geq 80\%$            | 82.0% on the 885-student test set (Logistic Regression)                    | ✓             |
| Precision (flags that are correct) $\geq 70\%$  | 78.7%                                                                      | ✓             |
| AUC-ROC $\geq 0.85$                             | 0.919                                                                      | ✓             |
| Cross-validation stability: AUC SD $\leq 0.02$  | 0.902 $\pm$ 0.012                                                          | ✓             |
| Fairness: recall gap between groups $\leq 0.10$ | Age gap 0.205 at baseline, reduced to 0.117 with group-specific thresholds | X Not yet met |

Table E.1 — Headline results (details in Chapters 4 and 5)

### Key findings

- Academic progress is the dominant risk driver. The 2nd-semester and overall approval rates (engineered features) rank first in every importance measure, followed by tuition-fee status and age at enrolment.
- Three of five models meet every performance target. Logistic Regression is selected for production because it meets every target at the default cut-off, explains itself through its coefficients, and trails the best ensembles by only about one AUC point, within cross-validation variability. XGBoost (precision 83.5% at the same 82.0% recall) is kept as the challenger.
- Students fall into natural risk tiers. Without using the outcome, K-Means finds three segments with dropout rates of 10.8%, 66.7% and 80.5%. This independently confirms the supervised signal and gives a simple basis for planning advisor capacity.
- Flag rates differ by group because real risk differs, but detection is uneven. The model catches only 71.6% of dropouts aged 20 or under and 62.1% of dropouts holding scholarships. Group-specific thresholds raise detection of young dropouts to 80.4% and overall recall to 85.9%, while keeping precision at 72.4%.

### Recommendations

- Deploy the Logistic Regression model as a mid-year review tool, with an advisor reviewing every flag. Use flags only to offer support, never to sanction.
- Adopt group-specific thresholds (cut-off 0.311 for students aged 20 or under and for scholarship holders) once the institution's governance or ethics body has approved them.
- Run a pilot to measure the retention lift. At an assumed 20% intervention success rate, about 51 extra students per 1,000 would stay enrolled; this assumption must be tested.

4. Monitor performance and fairness every term, and use a 1st-semester model (AUC 0.907) for earlier, lighter-touch screening.

# About this report

This report brings together the five written deliverables of the capstone in one document. Each chapter corresponds to one capstone step, and the chapters follow the order in which the work was done. Each chapter builds on the decisions of the one before:

| Chapter                                              | Capstone step | Question answered                                                         |
|------------------------------------------------------|---------------|---------------------------------------------------------------------------|
| 1. Problem Understanding and Framing                 | Step 1        | What should be predicted, for whom, when, and how will success be judged? |
| 2. Dataset Overview and Data Dictionary              | Step 2        | Is the data fit for purpose, and what does every variable mean?           |
| 3. Exploratory Data Analysis and Feature Engineering | Step 3        | What drives dropout, and which features should the models use?            |
| 4. Model Implementation and Comparison               | Step 4        | Which model best meets the success targets, and why?                      |
| 5. Explainability, Bias and Fairness Analysis        | Step 5        | How does the model decide, where are its limits, and is it fair?          |
| 6. Conclusions and Recommendations                   | —             | What should the institution do next?                                      |

Table 0.1 — Report structure

## Reproducibility

Every number in this report is produced by the code in the project's GitHub repository. After "pip install -r requirements.txt", the single command "python run\_all.py" runs the nine pipeline scripts in order. All randomness is fixed (random\_state = 42), all paths are relative to the project folder, and two full runs produce identical results. Appendix B lists the scripts and the outputs each one produces.

## Companion deliverables

- **GitHub repository:** source code, data, saved models, figures and the results-overview notebook (notebooks/01\_results\_overview.ipynb).
- **Technical presentation:** a 12-slide deck for a data-science audience covering method, results, explainability and fairness.
- **Business presentation:** a 10-slide deck for institutional decision-makers covering the business case, recommendations and pilot plan.

## Conventions

- Tables and figures are numbered by chapter (Table 3.2 is the second table in Chapter 3).
- Unless stated otherwise, 'recall' and 'precision' refer to the Dropout class on the held-out 885-student test set.
- Script names refer to files in the repository's src/ folder.

# 1. Problem Understanding and Framing

## Problem statement

Almost one in three students (32.1%) at an educational institution drops out before completing their degree. Advisors cannot give every student intensive support, and today they usually learn who is struggling only after a student has already disengaged.

Objective: Build a model that, using information available by the end of a student's first year, estimates each student's probability of dropping out. The model should flag at-risk students early enough in a 3–4-year degree for advisors to intervene. It must catch at least 80% of future dropouts (recall), keep at least 70% of flags correct (precision) and rank students reliably (AUC-ROC  $\geq 0.85$ ). It must also be explainable to advisors and fair across student groups.

Task type: supervised binary classification (Dropout vs not Dropout), supported by clustering for student segmentation.

## 1.1 Business context

### Why dropout matters

| Stakeholder           | Cost of a dropout                                                                                                                              | What they need from this project                                       |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Students and families | Lost time and tuition, debt without a qualification, lower lifetime earnings and confidence                                                    | Help offered before problems become irreversible                       |
| Institution           | Lost tuition revenue for the remaining years; wasted teaching and admission costs; lower completion rates, which affect reputation and funding | Higher retention and better use of limited advisor and support budgets |
| Academic advisors     | Effort spread thinly, or arriving too late                                                                                                     | A short, reliable, explained list of whom to contact first             |
| Society and funders   | Public investment in places that do not produce graduates                                                                                      | Evidence that support resources are targeted and fair                  |

Table 1.1 — Stakeholders and costs of dropout

### The decision this project supports

The institution must decide, after the first year's results are known, which students to offer proactive support such as academic tutoring, mentoring, flexible study plans or financial-aid counselling. Support capacity is limited, so the decision is in effect a ranking: whom to contact first. Two kinds of error have very different costs:

- Missing a student who will drop out (false negative): the student receives no help and is likely lost. This is the expensive error: lost tuition, an incomplete degree, and harm to the student.
- Flagging a student who would have stayed (false positive): an advisor spends time on a supportive conversation that was not strictly needed. This is the cheaper error, with a small cost in advisor time and little harm to the student, provided the support is non-punitive.

This asymmetry drives every metric choice in Section 1.4: the model should favour catching dropouts (recall), while keeping enough precision that advisor workload stays manageable.

### Scope and constraints

- Supportive use only: predictions trigger offers of help, never sanctions, admissions or scholarship decisions.
- Human in the loop: advisors review every flag and make the final decision.
- Explainability: each flag must come with understandable reasons, such as 'low 2nd-semester approval rate' or 'tuition fees in arrears'.

- Fairness: the model must not systematically miss at-risk students in particular groups (gender, age, socioeconomic status).
- Data protection: anonymised data, minimal access, and compliance with applicable law.

## 1.2 Data science framing

| Element          | Definition                                                                                                                                   |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Unit of analysis | One student (one row per student)                                                                                                            |
| Prediction point | End of the student's first academic year (after 1st- and 2nd-semester results), in a 3–4-year degree                                         |
| Inputs (X)       | 36 variables known by that point: demographic, socioeconomic, macro-economic, enrolment data, and 1st- and 2nd-semester academic performance |
| Target (y)       | Dropout = 1 if the student's final status is Dropout; 0 if Graduate or still Enrolled                                                        |
| Model output     | A probability of dropout, from 0 to 1, for each student                                                                                      |
| Decision rule    | Flag a student when the probability is at or above a cut-off; the cut-off is chosen on validation data to meet the recall target             |
| Data             | Public UCI dataset: 4,424 students; 1,421 dropouts (32.1%); class ratio 2.11 : 1                                                             |

Table 1.2 — Problem formulation

**Why 'Dropout vs the rest'.** The source data has three outcomes: Graduate, Dropout and Enrolled. The business decision is binary (offer support or not), and the group to find is dropouts. 'Enrolled' students are still studying at the end of the normal course duration. They have not left, so they belong with Graduate on the 'stay' side. Merging the two gives one clear yes/no question and a larger minority class to learn from.

## 1.3 Task type and justification

The problem is a **supervised binary classification** task. The other task types were considered and rejected as the main approach:

| Task type                  | Fit            | Reasoning                                                                                                                                                                                                           |
|----------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Binary classification      | ✓ Primary      | The target is a known yes/no label (dropped out or not), recorded for every student in historical data, so the model learns from labelled examples. The predicted probability also gives the ranking advisors need. |
| Multi-class classification | Considered     | Predicting all three outcomes is possible, but the decision is binary, and 'Enrolled' is an unfinished outcome that adds noise without changing the action taken.                                                   |
| Regression                 | ✗              | There is no continuous target to predict, such as time to dropout or final grade.                                                                                                                                   |
| Recommendation             | ✗              | There is no record of which interventions students received or how they responded, so there is nothing to learn recommendations from.                                                                               |
| Anomaly detection          | ✗ as main task | Dropout is not rare (32.1% of students), so it is a common class to predict, not an anomaly. Isolation Forest is used later only as an extra check for unusual profiles.                                            |
| Clustering                 | ✓ Supporting   | Unsupervised segmentation finds natural student groups (e.g. low, medium and high risk tiers) to help plan support capacity and interpret the model. It is not a substitute for labelled prediction.                |

Table 1.3 — Candidate task types

## 1.4 Success metrics (model)

Targets are set for the **Dropout class** on a held-out **test set** the model never saw during training or tuning:

| Metric                         | Target         | Definition                                                                            | Why this metric                                                                                     |
|--------------------------------|----------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Recall (sensitivity) — PRIMARY | ≥ 80%          | Share of actual dropouts the model flags                                              | Missing a dropout is the costliest error; at most 1 in 5 dropouts should be missed                  |
| Precision                      | ≥ 70%          | Share of flagged students who actually drop out                                       | Keeps advisor workload credible: at least 7 in 10 contacts are truly at risk                        |
| AUC-ROC                        | ≥ 0.85         | Probability that a random dropout is ranked above a random non-dropout                | Measures ranking quality regardless of cut-off, for fair model comparison and prioritising students |
| F1 score                       | Report         | Harmonic mean of recall and precision                                                 | A single balanced summary when comparing models                                                     |
| Cross-validation stability     | AUC SD ≤ 0.02  | Spread of AUC across 5 stratified folds                                               | Shows the result is reliable, not a lucky split                                                     |
| Fairness (equal opportunity)   | TPR gap ≤ 0.10 | Largest difference in recall between groups (gender, age, scholarship, debtor status) | At-risk students in every group should have a similar chance of being offered help                  |
| Accuracy                       | Report only    | Share of all predictions that are correct                                             | Not a target: a model that never predicts dropout already scores 67.9%                              |
| Silhouette score (clustering)  | Report         | How well-separated the clusters are (–1 to 1)                                         | Judges whether student segments are distinct enough to be useful                                    |

Table 1.4 — Model success metrics and targets

## Baselines the model must beat

| Naive approach                    | Recall               | Precision | Accuracy | Lesson                                                                 |
|-----------------------------------|----------------------|-----------|----------|------------------------------------------------------------------------|
| Predict 'no dropout' for everyone | 0%                   | —         | 67.9%    | High accuracy with zero value — accuracy cannot be the target          |
| Flag every student                | 100%                 | 32.1%     | 32.1%    | Catches everyone but overwhelms advisors                               |
| Flag students at random           | Equals the flag rate | 32.1%     | ≈ 56%*   | Precision stays at the base rate — the model must beat 32.1% precision |

Table 1.5 — Naive baselines (\*when flagging the same 32% share of students at random)

**Why these thresholds.** Recall ≥ 80% reflects how expensive a missed dropout is. Precision ≥ 70% is more than double the 32.1% chance level, while staying achievable together with high recall. AUC ≥ 0.85 is a widely used marker of strong discrimination for early-warning models. The fairness target makes 'success' include not leaving any group behind.

## 1.5 Business KPIs

Model metrics show the model works; business KPIs show the intervention programme works. They are measured during deployment, ideally through a pilot that compares supported students with a comparable group.

| KPI                        | Formula                                                                                                                   | Target                                  | Owner            |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------|
| Retention lift             | (Retention rate of flagged students who received support – retention rate of a comparable unsupported group) ÷ the latter | +5% to +10% relative                    | Student services |
| Institutional dropout rate | Dropouts ÷ cohort size, tracked per cohort                                                                                | Measurable fall from the 32.1% baseline | Management       |
| Revenue protected          | Additional students retained × annual tuition × average remaining years                                                   | Exceeds programme cost (ROI > 1)        | Finance          |
| Cost per retained student  | Total programme cost ÷ additional students retained                                                                       | Track and reduce year on year           | Finance          |

| KPI                  | Formula                                                                               | Target                              | Owner      |
|----------------------|---------------------------------------------------------------------------------------|-------------------------------------|------------|
| Advisor efficiency   | Share of advisor contacts that reach truly at-risk students (= precision in practice) | $\geq 70\%$                         | Advisors   |
| Time to intervention | Days from semester results to first advisor contact                                   | $\leq 14$ days                      | Advisors   |
| Equitable reach      | Share of at-risk students contacted, by group                                         | Gap $\leq 10$ points between groups | Governance |

Table 1.6 — Business KPIs

### How model metrics translate into business value (illustration)

For every **1,000 students**, about **321** would drop out without help. At the target recall of 80%, the model flags about **257** of them. At 70% precision, it also flags about **110** students who would have stayed, about 367 flags in total.

If support then keeps an **assumed 20%** of the flagged true dropouts enrolled, about **51 students** are retained per 1,000. That would cut the dropout rate by about **5 percentage points** (32.1% → about 27%). The 20% success rate is an **assumption to be tested in a pilot**, not a measured result. The revenue-protected KPI turns retained students into money using the institution's actual tuition and programme costs.

## 1.6 Capstone linkage

The Module 1 output — this framing — sets the agenda for Chapters 1–3 and the criteria used to judge every later step:

| Module 1 output                                         | Feeds                             | How it is used                                                                                               |
|---------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Problem statement and decision                          | Step 1 → Chapter 1 (this chapter) | Defines what is predicted, for whom, when, and why                                                           |
| Prediction point and data needs                         | Step 2 → Chapter 2                | Selects a dataset with labelled outcomes and first-year data; data dictionary and quality checks             |
| Target definition (Dropout vs rest) and class imbalance | Step 3 → Chapter 3                | Binary target, stratified splits, class weights; engineered features capture first-year academic progress    |
| Success metrics and targets                             | Steps 4–5 → Chapters 4–5          | Recall / precision / AUC targets select the cut-off and the model; the fairness target drives the bias audit |
| Business KPIs                                           | Step 6 → presentations; Chapter 6 | Basis for the business case, pilot design and monitoring dashboard                                           |

Table 1.7 — Linkage between Module 1 and the capstone steps

### Traceability: targets and results achieved

| Target (set here)             | Result in Chapters 4–5 (selected model: Logistic Regression)                                  | Met?                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------|
| Recall $\geq 80\%$            | 82.0% on the test set                                                                         | ✓                                                  |
| Precision $\geq 70\%$         | 78.7%                                                                                         | ✓                                                  |
| AUC-ROC $\geq 0.85$           | 0.919                                                                                         | ✓                                                  |
| CV AUC SD $\leq 0.02$         | 0.902 $\pm$ 0.012                                                                             | ✓                                                  |
| Fairness: TPR gap $\leq 0.10$ | Baseline gap of 0.205 for age (Chapter 5 audit model); reduced to 0.117 with group thresholds | X Not yet met — mitigation and monitoring in place |

Table 1.8 — Results against the Chapter 1 targets (detailed in Chapters 4 and 5)

The three performance targets were met. The fairness target was not fully met: young students' dropouts are still under-detected even after mitigation. Chapter 5 reports this openly and sets out the safeguards and monitoring that respond to it.

The rest of the report follows the capstone steps in order. Chapter 2 establishes that the data can support this framing. Chapter 3 turns the data into model-ready features. Chapter 4 selects the model against the targets above, and Chapter 5 tests whether it can be trusted and is fair.

## 2. Dataset Overview and Data Dictionary

Chapter overview. The project uses the public, peer-reviewed Predict Students' Dropout and Academic Success dataset from the UCI Machine Learning Repository (CC BY 4.0). It holds real, anonymised records of 4,424 students across 17 undergraduate degrees at the Polytechnic Institute of Portalegre, Portugal, enrolled from 2008/09 to 2018/19. The 36 input variables cover demographic, socioeconomic, macro-economic, enrolment and academic-performance data, and the labelled outcome is Graduate, Dropout or Enrolled. The data is 100% complete with no duplicates, and every value falls within its valid range. It suits an early-warning dropout model directly. Its main limitations are a single institution, integer-coded categories, no ethnicity data, and a 2.11 : 1 class imbalance; all are documented and handled in later steps.

### 2.1 Data source and collection

| Item                    | Detail                                                                                                                                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset                 | Predict Students' Dropout and Academic Success                                                                                                                                                             |
| Repository              | UCI Machine Learning Repository, dataset 697 — DOI 10.24432/C5MC89                                                                                                                                         |
| Creators                | Realinho, V., Vieira Martins, M., Machado, J. & Baptista, L.                                                                                                                                               |
| Documentation           | Peer-reviewed data descriptor: Realinho et al. (2022), Data, 7(11), 146                                                                                                                                    |
| Licence                 | Creative Commons Attribution 4.0 (CC BY 4.0) — free use with attribution                                                                                                                                   |
| Institution and period  | Polytechnic Institute of Portalegre, Portugal; students enrolled 2008/09 to 2018/19; 17 undergraduate degrees (agronomy, design, education, nursing, journalism, management, social service, technologies) |
| Original sources        | Academic Management System; Support System for Teaching Activity; General Directorate of Higher Education (national admissions data); PORDATA (macro-economic indicators)                                  |
| Creators' preprocessing | Records merged from the four sources; duplicates and irrelevant enrolment types removed; anomalies and outliers treated; personal identifiers removed                                                      |
| Funding                 | SATDAP — Capacitação da Administração Pública (grant POCI-05-5762-FSE-000191), Portugal                                                                                                                    |
| Project file            | data/raw/students_dropout_academic_success.csv — a CSV copy of the UCI data (the outcome column is named 'target'); 4,424 rows × 37 columns                                                                |

Table 2.1 — Dataset provenance

### Why this dataset is fit for purpose

| Quality criterion          | Evidence                                                                                                                         |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Relevant to the problem    | Built specifically to 'identify students at risk at an early stage of their academic path', which is exactly this project's goal |
| Real and authoritative     | Drawn from an institution's own administrative systems and national admissions data, not synthetic or scraped                    |
| Credible and documented    | Published with a peer-reviewed data descriptor and a DOI; widely used as a benchmark for dropout prediction                      |
| Labelled outcome           | Every student's final status (Graduate / Dropout / Enrolled) is recorded at the end of the normal course duration                |
| Rich, multi-angle features | 36 variables across five domains (Section 2.2), so the model can weigh academic, financial and personal factors together         |
| Adequate size              | 4,424 students, including 1,421 dropouts — enough for training, validation and test splits and 5-fold cross-validation           |
| High data quality          | No missing values, no duplicate rows, all values within valid ranges (Section 2.3)                                               |
| Legal and ethical to use   | Open CC BY 4.0 licence; anonymised, with no names, IDs or contact details                                                        |

Table 2.2 — Justification of dataset quality

## 2.2 Dataset overview

| Property        | Value                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Records         | 4,424 students (one row per student)                                                                            |
| Columns         | 37 = 36 input variables + 1 outcome ("target")                                                                  |
| Storage types   | 29 integer, 7 decimal, 1 text                                                                                   |
| Outcome classes | Graduate 2,209 (49.9%); Dropout 1,421 (32.1%); Enrolled 794 (17.9%)                                             |
| Project target  | Dropout (1) vs not Dropout (0): 32.1% vs 67.9%, a 2.11 : 1 imbalance                                            |
| Typical student | Median age at enrolment 20; 64.8% female; 89.1% daytime attendance; 97.5% Portuguese; 24.8% scholarship holders |

Table 2.3 — Key facts

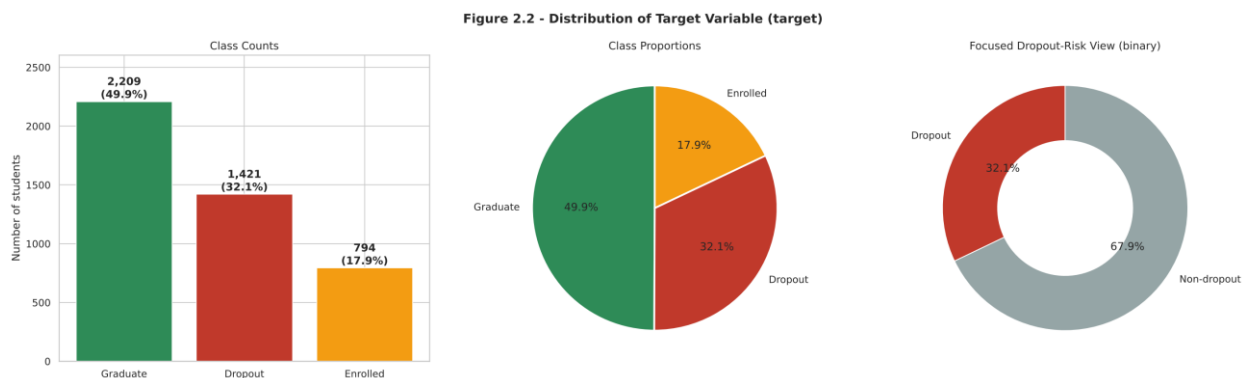


Figure 2.1 — Outcome distribution (src/01\_data\_quality\_eda.py)

### Feature types

| Measurement type                    | Count | Variables                                                                                                                                            |
|-------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Binary (0/1)                        | 8     | Daytime/evening attendance, Displaced, Educational special needs, Debtor, Tuition fees up to date, Gender, Scholarship holder, International         |
| Nominal categorical (integer codes) | 9     | Marital status, Application mode, Course, Previous qualification, Nationality, Mother's and Father's qualification, Mother's and Father's occupation |
| Ordinal                             | 1     | Application order (preference rank of the chosen course)                                                                                             |
| Discrete count                      | 11    | Age at enrolment; curricular units credited, enrolled, evaluations, approved and without evaluations (each for semesters 1 and 2)                    |
| Continuous                          | 7     | Previous qualification grade, Admission grade, 1st- and 2nd-semester average grade, Unemployment rate, Inflation rate, GDP                           |
| Outcome (categorical text)          | 1     | target: Graduate / Dropout / Enrolled                                                                                                                |

Table 2.4 — Variables by measurement type

| Domain                  | Count | Variables                                                                                                               | Known when?                                         |
|-------------------------|-------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Demographic             | 6     | Marital status, Nationality, Displaced, Gender, Age at enrolment, International                                         | At enrolment                                        |
| Socioeconomic           | 8     | Parents' qualifications and occupations, Educational special needs, Debtor, Tuition fees up to date, Scholarship holder | At enrolment (fees and debt change during the year) |
| Macro-economic          | 3     | Unemployment rate, Inflation rate, GDP (national values for the enrolment year)                                         | At enrolment                                        |
| Academic — at enrolment | 7     | Application mode, Application order, Course, Attendance regime, Previous qualification and its grade, Admission grade   | At enrolment                                        |

| Domain                 | Count | Variables                                                                                                          | Known when?                    |
|------------------------|-------|--------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Academic — performance | 12    | Curricular units credited / enrolled / evaluations / approved / grade / without evaluations, for semesters 1 and 2 | End of semester 1 / semester 2 |

Table 2.5 — Variables by domain and timing (timing matters for an early-warning model)

## 2.3 Data quality: missing values, duplicates and outliers

| Check                     | Result                                                                                                                            | Implication                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Missing values            | 0 of 163,688 cells — 100% complete                                                                                                | No imputation needed                                                                                              |
| Duplicate rows            | 0                                                                                                                                 | No de-duplication needed                                                                                          |
| Valid ranges              | All values within valid ranges: grades 95–190 on a 0–200 scale, average grades 0–18.9 on a 0–20 scale, age 17–70, counts $\geq 0$ | No invalid values to correct                                                                                      |
| Outliers (1.5 × IQR rule) | Most affected: 2nd-semester grade 19.8%, 1st-semester grade 16.4%, age at enrolment 10.0%, 1st-semester units enrolled 9.6%       | Genuine values, kept: zero grades identify students who passed or sat nothing; older students are a real subgroup |
| Zero inflation            | 1st-semester grade = 0 for 16.2% of students; 2nd-semester grade = 0 for 19.7%                                                    | A meaningful at-risk signal, not missing data                                                                     |
| Skewness                  | Age at enrolment +2.05; 1st-semester units enrolled +1.62; 1st-semester grade –1.57                                               | Handled by binning and scaling in Chapter 3                                                                       |
| Near-constant variables   | Educational special needs: 98.8% = 0; International: 97.5% = 0                                                                    | Low information value; kept, but did not reach the final feature set                                              |
| Rare categories           | e.g. Course 33 (12 students); widower (4); Application order 0 and 9 (1 student each)                                             | Too few for reliable group estimates                                                                              |
| Class imbalance           | Dropout 32.1% vs 67.9% (2.11 : 1)                                                                                                 | Stratified splits and class weighting in all models                                                               |

Table 2.6 — Data-quality profile (reproduced by `src/01_data_quality_eda.py`; the full tables are exported to Excel)

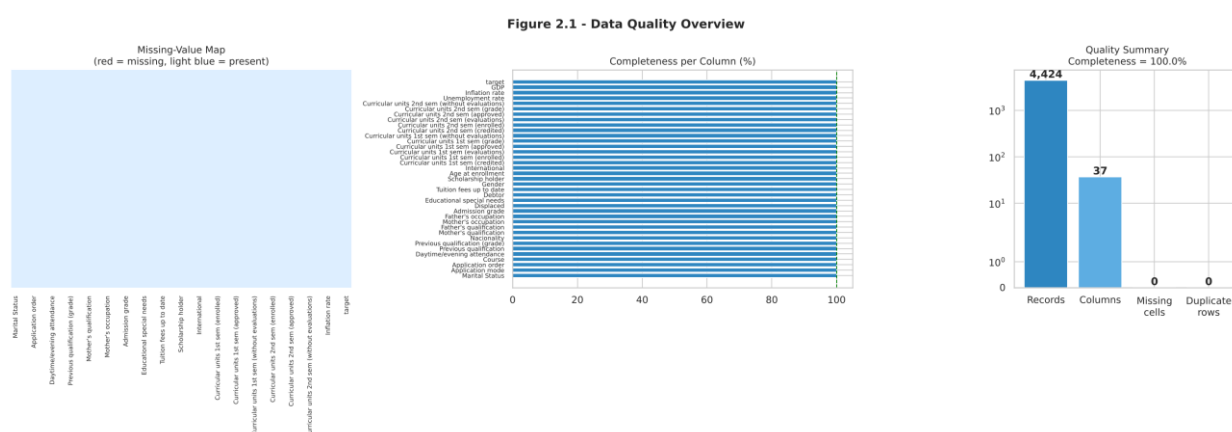


Figure 2.2 — Missing-value map, completeness per column and quality summary

### Observations worth noting

- Application order is documented as 0 (first choice) to 9 (last choice), but in practice almost every student has 1–6. 68.4% have 1, which is treated as first choice; values 0 and 9 occur once each.
- Parents' codes: mother's and father's qualification and occupation use the same coding scheme, but different subsets appear (29 vs 34 qualification codes; 32 vs 46 occupation codes).
- Large numeric codes: Course and occupation codes are large integers (e.g. 9500 = Nursing). They are labels, not quantities, and should not be read as amounts.
- Cohort-level values: the macro-economic variables take only 9–10 distinct values each, because they describe the year of enrolment rather than the individual student.

## 2.4 Data dictionary

All 37 columns, in file order. Observed ranges come from the project data; code meanings come from the creators' documentation (UCI page and Realinho et al., 2022). Full code lists for the nominal variables are in the Appendix.

| #  | Variable (exact column name)                   | Type       | Unit / scale  | Allowed / observed values     | Description                                                                                                 |
|----|------------------------------------------------|------------|---------------|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1  | Marital Status                                 | Nominal    | Code          | 1–6 (6 codes)                 | Marital status at enrolment: 1 single, 2 married, 3 widower, 4 divorced, 5 facto union, 6 legally separated |
| 2  | Application mode                               | Nominal    | Code          | 18 codes, 1–57 (Table A1)     | Admission route, e.g. 1st-phase general contingent, over-23 entry, transfer                                 |
| 3  | Application order                              | Ordinal    | Rank          | 0–9 (observed 0–6 and 9)      | Preference rank of the course in the student's application; 1 = first choice                                |
| 4  | Course                                         | Nominal    | Code          | 17 codes, 33–9991 (Table A2)  | Degree programme enrolled in                                                                                |
| 5  | Daytime/evening attendance                     | Binary     | Flag          | 1 daytime, 0 evening          | Attendance regime                                                                                           |
| 6  | Previous qualification                         | Nominal    | Code          | 17 codes, 1–43 (Table A3)     | Highest qualification before entry                                                                          |
| 7  | Previous qualification (grade)                 | Continuous | Points, 0–200 | Observed 95–190               | Grade of the previous qualification                                                                         |
| 8  | Nacionality                                    | Nominal    | Code          | 21 codes, 1–109 (Table A4)    | Nationality (column name spelled as in the source)                                                          |
| 9  | Mother's qualification                         | Nominal    | Code          | 1–44; 29 observed (Table A5)  | Mother's highest education level                                                                            |
| 10 | Father's qualification                         | Nominal    | Code          | 1–44; 34 observed (Table A5)  | Father's highest education level                                                                            |
| 11 | Mother's occupation                            | Nominal    | Code          | 0–194; 32 observed (Table A6) | Mother's occupation group                                                                                   |
| 12 | Father's occupation                            | Nominal    | Code          | 0–195; 46 observed (Table A6) | Father's occupation group                                                                                   |
| 13 | Admission grade                                | Continuous | Points, 0–200 | Observed 95–190               | Score used for admission to the course                                                                      |
| 14 | Displaced                                      | Binary     | Flag          | 1 yes, 0 no                   | Student lives away from home to study                                                                       |
| 15 | Educational special needs                      | Binary     | Flag          | 1 yes, 0 no                   | Has recognised special educational needs                                                                    |
| 16 | Debtor                                         | Binary     | Flag          | 1 yes, 0 no                   | Owes money to the institution                                                                               |
| 17 | Tuition fees up to date                        | Binary     | Flag          | 1 yes, 0 no                   | Tuition payments are current                                                                                |
| 18 | Gender                                         | Binary     | Flag          | 1 male, 0 female              | Student's gender                                                                                            |
| 19 | Scholarship holder                             | Binary     | Flag          | 1 yes, 0 no                   | Receives a scholarship                                                                                      |
| 20 | Age at enrollment                              | Discrete   | Years         | 17–70                         | Age when first enrolled                                                                                     |
| 21 | International                                  | Binary     | Flag          | 1 yes, 0 no                   | International student                                                                                       |
| 22 | Curricular units 1st sem (credited)            | Count      | Units         | 0–20                          | Units credited from prior study, semester 1                                                                 |
| 23 | Curricular units 1st sem (enrolled)            | Count      | Units         | 0–26                          | Units enrolled, semester 1                                                                                  |
| 24 | Curricular units 1st sem (evaluations)         | Count      | Assessments   | 0–45                          | Number of evaluations taken, semester 1                                                                     |
| 25 | Curricular units 1st sem (approved)            | Count      | Units         | 0–26                          | Units passed, semester 1                                                                                    |
| 26 | Curricular units 1st sem (grade)               | Continuous | Points, 0–20  | 0–18.9                        | Average grade in semester 1 (0 if nothing graded)                                                           |
| 27 | Curricular units 1st sem (without evaluations) | Count      | Units         | 0–12                          | Enrolled units with no evaluation, semester 1                                                               |

| #  | Variable (exact column name)                   | Type              | Unit / scale | Allowed / observed values   | Description                                                                             |
|----|------------------------------------------------|-------------------|--------------|-----------------------------|-----------------------------------------------------------------------------------------|
| 28 | Curricular units 2nd sem (credited)            | Count             | Units        | 0–19                        | Units credited from prior study, semester 2                                             |
| 29 | Curricular units 2nd sem (enrolled)            | Count             | Units        | 0–23                        | Units enrolled, semester 2                                                              |
| 30 | Curricular units 2nd sem (evaluations)         | Count             | Assessments  | 0–33                        | Number of evaluations taken, semester 2                                                 |
| 31 | Curricular units 2nd sem (approved)            | Count             | Units        | 0–20                        | Units passed, semester 2                                                                |
| 32 | Curricular units 2nd sem (grade)               | Continuous        | Points, 0–20 | 0–18.6                      | Average grade in semester 2 (0 if nothing graded)                                       |
| 33 | Curricular units 2nd sem (without evaluations) | Count             | Units        | 0–12                        | Enrolled units with no evaluation, semester 2                                           |
| 34 | Unemployment rate                              | Continuous        | %            | 7.6–16.2 (10 values)        | National unemployment rate in the enrolment year                                        |
| 35 | Inflation rate                                 | Continuous        | %            | –0.8–3.7 (9 values)         | National inflation rate in the enrolment year                                           |
| 36 | GDP                                            | Continuous        | % growth     | –4.06–3.51 (10 values)      | National GDP growth in the enrolment year                                               |
| 37 | target                                         | Nominal (outcome) | Label        | Graduate, Dropout, Enrolled | Status at the end of the normal course duration; the project models Dropout vs the rest |

Table 2.7 — Data dictionary (36 inputs + outcome). No column has missing values.

## 2.5 Limitations and how later steps address them

| Limitation                         | Why it matters                                                     | Addressed in                                                                             |
|------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Single institution, one country    | Patterns may not transfer to other universities or countries       | Stated as a limitation; recommend revalidating before use elsewhere (Chapters 4–5)       |
| Categories stored as integer codes | Codes such as Course = 9500 can be mistaken for quantities         | Kept as labels; one-hot encoding recommended (Chapter 3)                                 |
| Semester 2 data arrives mid-year   | The strongest predictors are not available at enrolment            | Timing analysis; earlier-stage models compared (Chapter 5 ablation)                      |
| Class imbalance (2.11 : 1)         | Models could under-detect dropouts                                 | Stratified splits, class weights and validation-chosen cut-offs (Chapters 3–4)           |
| No race or ethnicity data          | Fairness cannot be audited for these groups                        | Audit uses gender, age, scholarship, debtor and international status instead (Chapter 5) |
| 'Enrolled' is an open outcome      | Some still-enrolled students may later drop out or graduate        | Merged with Graduate as 'not dropout'; noted as a limitation                             |
| Data ends in 2018/19               | Recent changes (e.g. post-2020 learning modes) are not represented | Monitoring and periodic retraining recommended (Chapter 5)                               |

Table 2.8 — Dataset limitations

The dataset is complete, credible and directly relevant to the early-warning task. Its limitations are known and each has a planned response. Chapter 3 takes this data through cleaning, applied exploratory analysis and feature engineering to produce the feature set used by every model.

## 3. Exploratory Data Analysis and Feature Engineering

**Chapter overview.** The dataset of 4,424 students is complete: no missing values, no duplicates and no implausible values. Statistical outliers were kept because they are genuine students, not errors. EDA shows that dropout is driven mainly by academic progress, tuition-fee status, age and financial circumstances. Twenty domain-derived features were created; they take the top two places in every ranking and fill 14 of the 25 selected features. Three feature-selection methods (filter, embedded and wrapper) agree on a core of 13 features. The final set of 25 features carries 81% of total Random Forest importance. PCA shows that 7 components capture 80% of the variance of the selected features. It is used for visualisation and clustering checks rather than modelling, so the features stay interpretable. Every number in this report is reproduced by the code in the project repository.

### 3.0 Reproducible code

| Script (src/)                 | What it does                                                                                       | Outputs                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 01_data_quality_eda.py        | Data quality checks, distributions, outliers, plausibility, categorical analysis                   | 8 charts and Excel tables in reports/figures/section2_eda/ |
| 02_eda_feature_engineering.py | Cleaning summary, correlations, 22 engineered features, three selection methods, PCA               | data/processed/feature_list.txt                            |
| 08_unsupervised_analysis.py   | Clustering tendency (silhouette), K-Means, PCA visualisation, dendrogram                           | reports/figures/unsupervised/                              |
| 03–07_train_*.py              | Apply the same engineering, then scaling and the train / validation / test split inside each model | models/, reports/model_metrics.csv                         |

Table 3.1 — Chapter 3 scripts. Run all with "python run\_all.py"; every path is relative to the project folder, so the code runs on any computer.

### 3.1 Data cleaning

| Check                 | Result                                                                                                                  | Decision and justification                                                          |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Missing values        | 0 of 163,688 cells (100% complete)                                                                                      | No imputation needed                                                                |
| Duplicate rows        | 0                                                                                                                       | No removal needed                                                                   |
| Data types            | 29 integer, 7 decimal, 1 text (target)                                                                                  | Categorical variables are already integer-coded in the source; no type fixes needed |
| Plausibility          | All 12 key numeric variables within valid ranges (grades 0–20 and 95–190, age 17–70, rates within real economic bounds) | No values corrected or removed                                                      |
| Outliers (IQR rule)   | 28 columns contain values beyond $1.5 \times$ IQR; highest shares: 2nd-sem grade 19.8%, 1st-sem grade 16.4%, age 10.0%  | Retained — see below                                                                |
| Near-constant columns | Educational special needs (98.8% one value), International (97.5%)                                                      | Kept in the data, but low information; neither reached the final feature set        |

Table 3.2 — Data-quality results (src/01\_data\_quality\_eda.py, src/02\_eda\_feature\_engineering.py)

#### Why outliers were retained

- They are real, not errors. Every value passed the plausibility check: no negative grades, impossible ages or out-of-range scores.
- The 'outliers' carry the signal. 16.2% of students have a 1st-semester grade of 0 and 19.7% a 2nd-semester grade of 0. These are students who were not evaluated or passed nothing — exactly the at-risk group. Capping or removing them would erase the strongest evidence of dropout.

- Many flagged columns are codes, not measurements. The IQR rule also flags integer category codes (e.g. course, application mode), where 'distance from the median' has no meaning.
- The models are robust to them. Tree models are unaffected by extreme values, and the linear models use standardised features with regularisation.

Skewness was noted but not transformed. Age at enrolment (skew 2.05) is handled by binning instead (Section 3.3). The heavy left skew of the grade variables comes from the zero-grade group, which is meaningful.

## 3.2 Applied exploratory data analysis

### Target distribution

The business question is whether a student will drop out, so the target is framed (Figure 2.1) as Dropout (32.1%) vs not Dropout (67.9%), a 2.11 : 1 imbalance. This ratio sets the class weight (2.11) used by every model, and it is why all data splits are stratified.

### Distributions and relationships with dropout

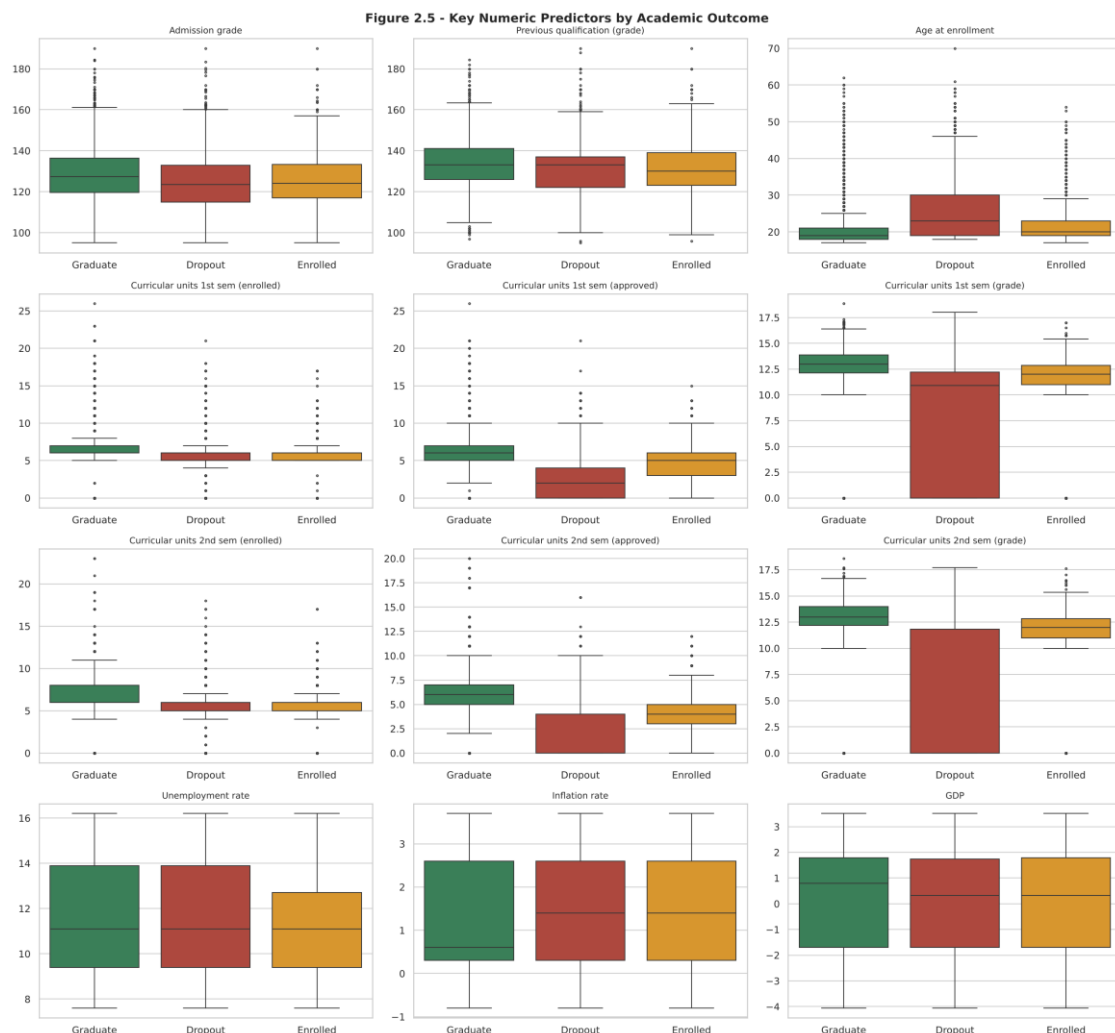


Figure 3.1 — Key numeric predictors by outcome (src/01\_data\_quality\_eda.py)

- Academic progress separates the groups most clearly. Dropouts' approved units and grades cluster at or near zero, while graduates sit tightly around 5–7 approved units and grades of about 12–14; still-enrolled students fall in between.
- Age: the median dropout enrolled at 23, against about 19 for graduates, with a long tail of older dropouts.

- Admission and previous-qualification grades differ only slightly between groups, so prior attainment matters far less than performance at university.
- Macro-economic variables (unemployment, inflation, GDP) are almost identical across outcomes. They describe the enrolment year, not the student.

| Variable                            | Correlation with dropout | Direction                        |
|-------------------------------------|--------------------------|----------------------------------|
| Curricular units 2nd sem (grade)    | -0.572                   | Higher grades → less dropout     |
| Curricular units 2nd sem (approved) | -0.570                   | More units passed → less dropout |
| Curricular units 1st sem (grade)    | -0.481                   | Higher grades → less dropout     |
| Curricular units 1st sem (approved) | -0.479                   | More units passed → less dropout |
| Tuition fees up to date             | -0.429                   | Fees paid → less dropout         |
| Age at enrollment                   | +0.254                   | Older → more dropout             |
| Scholarship holder                  | -0.245                   | Scholarship → less dropout       |
| Debtor                              | +0.229                   | In debt → more dropout           |
| Gender (1 = male)                   | +0.204                   | Male → more dropout              |

Table 3.3 — Strongest correlations with dropout (script output shows absolute values; direction confirmed from the data)

## Categorical and group patterns

| Group                       | Dropout rate | Compared with             |
|-----------------------------|--------------|---------------------------|
| Tuition fees NOT up to date | 86.6%        | 32.1% overall             |
| Debtor                      | 62.0%        | 32.1% overall             |
| Age > 25                    | 55.9%        | 24.8% for age ≤ 25        |
| Male                        | 45.1%        | 25.1% for female          |
| Scholarship holder          | 12.2%        | 38.7% without scholarship |

Table 3.4 — Dropout rate by group (src/01 and src/02)

These gaps are large and practically important. They point to financial strain and non-traditional (older) students as risk factors alongside academic performance. They also flag gender, age and scholarship status as sensitive attributes for the fairness audit in Chapter 5.

## Multicollinearity

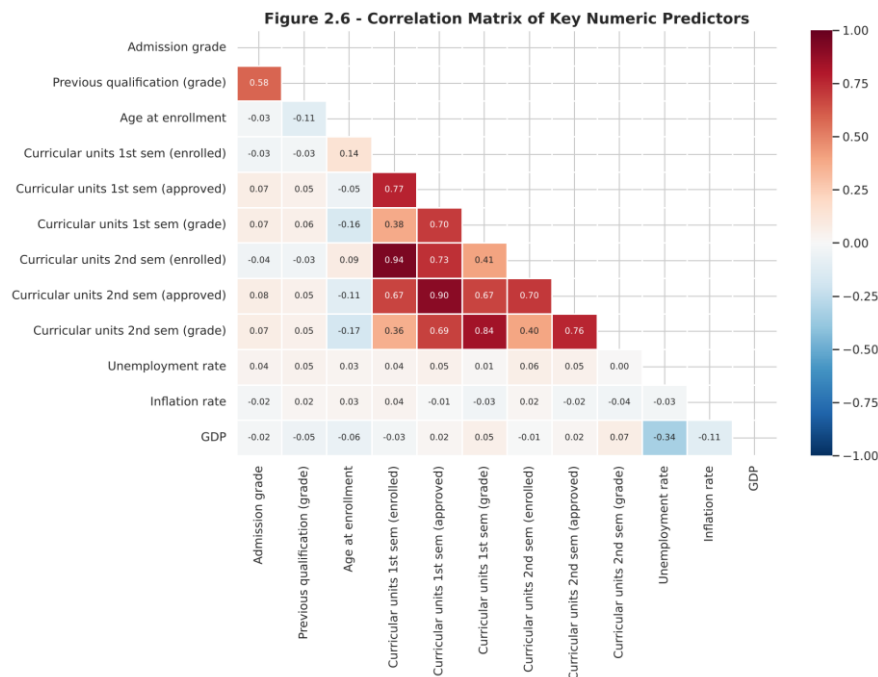


Figure 3.2 — Correlation matrix of key numeric predictors

1st- and 2nd-semester approved units are strongly correlated ( $r = 0.90$ ), and so are grades and approved units within a semester ( $r \approx 0.70$ ). Admission grade and previous-qualification grade correlate at 0.58. This redundancy motivated (a) ratio features that combine enrolled and approved units into one signal, (b) feature selection to drop near-duplicates, and (c) regularised or tree-based models that tolerate correlated inputs.

## Clustering tendency

Silhouette analysis on the 25 selected features (K-Means,  $K = 2-6$ ) gives scores of 0.411, 0.276, 0.287, 0.144 and 0.218. The data has **moderate natural cluster structure**, strongest at  $K = 2$ . That split separates academically engaged students from disengaged ones. The PCA projection (Figure 3.3) shows the same picture: a dense graduate-dominated region and a separate high-dropout region. Chapter 4 builds on this with a three-tier K-Means segmentation (dropout rates 10.8%, 66.7% and 80.5%).

## 3.3 Feature engineering

### Domain-derived features

Twenty-two features were created in five groups. Two of them (has\_special\_needs, displaced\_student) are renamed copies kept for readability, so 20 are genuinely new. Ratios add 1 to the denominator so that students with zero enrolled units do not cause division by zero.

| Group                   | Features                                                          | Definition                               | Rationale                                          |
|-------------------------|-------------------------------------------------------------------|------------------------------------------|----------------------------------------------------|
| Academic engagement (9) | approval_rate_1st,<br>approval_rate_2nd,<br>overall_approval_rate | Units approved ÷ (units enrolled + 1)    | Normalises for course load: 3 of 3 passed ≠ 3 of 8 |
|                         | engagement_1st,<br>engagement_2nd                                 | Evaluations ÷ (units enrolled + 1)       | Whether the student sits assessments at all        |
|                         | avg_grade_1st_2nd,<br>grade_decline                               | Mean of semester grades; 1st – 2nd grade | Overall level and deterioration over the year      |

| Group              | Features                                           | Definition                                                               | Rationale                                            |
|--------------------|----------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------|
|                    | total_approved,<br>disengagement_score             | Sum of approved units;<br>weighted count of units without<br>evaluations | Cumulative progress;<br>disengagement signal         |
| Financial (1)      | financial_hardship                                 | 2 × debtor + 1 × fees not up to<br>date (0–3)                            | Combines the two strongest<br>financial risk signals |
| Demographic (5)    | parental_education_avg,<br>parental_occupation_avg | Mean of mother's and father's<br>codes                                   | Family background in one<br>feature                  |
|                    | age_non_traditional                                | 1 if age at enrolment > 25                                               | Binning — captures the jump<br>in risk after 25      |
|                    | has_special_needs,<br>displaced_student            | Copies of original flags                                                 | Readable names (not new<br>information)              |
| Motivation (2)     | is_first_choice                                    | 1 if application order = 1                                               | Studying one's first-choice<br>course                |
|                    | admission_grade_zscore                             | (grade – mean) ÷ standard<br>deviation                                   | Standardised prior attainment                        |
| Clinical flags (5) | no_attempt_1st,<br>no_attempt_2nd                  | 1 if no evaluations in the<br>semester                                   | Early warning of total<br>disengagement              |
|                    | dropped_mid_year                                   | 1 if 0 units enrolled in the 2nd<br>semester                             | Effectively left after semester<br>1                 |
|                    | failed_majority_1st,<br>failed_majority_2nd        | 1 if approved < half of enrolled                                         | Rule an advisor could apply<br>directly              |

Table 3.5 — Engineered features (src/02\_eda\_feature\_engineering.py); the dataset grows from 37 to 60 columns

Evidence the engineering worked: the two highest-ranked features in both the filter and embedded methods are engineered (approval\_rate\_2nd, overall\_approval\_rate). Engineered features fill 14 of the 25 final features, and the engineered approval ratios load most heavily on the first principal component.

## Scaling

- StandardScaler (mean 0, standard deviation 1) for scale-sensitive models: Logistic Regression, SVM and the neural network.
- No data leakage: the scaler is fitted on the training set only and then applied to validation and test data, inside each model script.
- Tree models are scale-invariant; they receive the same scaled inputs for consistency, which does not change their splits.
- Clustering and PCA use features standardised across the whole dataset, because they are descriptive and make no predictions.

## Encoding

The source data already stores every categorical variable as integer codes: binary flags as 0/1, and nominal variables such as course, application mode and parents' occupation as category codes. Binary flags need no further encoding. Nominal codes were used as-is, which suits tree-based models but makes a linear model treat them as ordered numbers. Course does appear among the selected features, so **one-hot encoding of nominal variables is a recommended refinement** (see 3.7).

## Binning

- Age: a non-traditional student flag (> 25), justified by the jump in dropout from 24.8% to 55.9%; three age bands (≤ 20, 21–25, > 25) are used in the fairness audit.
- Academic thresholds: binary flags such as 'failed the majority of units' and 'no evaluations', which turn continuous measures into rules advisors can act on.
- Financial hardship: collapsed into a 0–3 ordinal scale.

### 3.4 Feature importance and explainability

Importance was measured three ways in Chapter 3, then confirmed with SHAP and LIME during modelling (Chapter 5).

| Rank | Random Forest importance                    | F-statistic (filter)                     | Mean  SHAP , XGBoost (Chapter 5) |
|------|---------------------------------------------|------------------------------------------|----------------------------------|
| 1    | approval_rate_2nd (0.109)                   | approval_rate_2nd (3,403)                | overall_approval_rate (0.973)    |
| 2    | overall_approval_rate (0.092)               | overall_approval_rate (3,187)            | approval_rate_2nd (0.503)        |
| 3    | Curricular units 2nd sem (approved) (0.065) | failed_majority_2nd (2,792)              | Tuition fees up to date (0.411)  |
| 4    | approval_rate_1st (0.058)                   | approval_rate_1st (2,377)                | Age at enrollment (0.284)        |
| 5    | total_approved (0.050)                      | Curricular units 2nd sem (grade) (2,148) | Course (0.283)                   |

Table 3.6 — Top-5 features under three importance measures

All three measures agree that academic progress, captured best by the engineered approval-rate features, is the dominant driver, followed by tuition-fee status and age. SHAP adds the direction of each effect: higher approval rates and fees paid lower risk, while older age raises it. LIME confirms the same reasons for individual students. The full explainability analysis (SHAP, LIME, PDP, ICE) is in Chapter 5.

### 3.5 Feature selection

All three families of selection method were applied to the 58 numeric features, so that no single method's bias decides the result:

| Method                          | Type     | How it works                                                                                                  | Selected |
|---------------------------------|----------|---------------------------------------------------------------------------------------------------------------|----------|
| ANOVA F-statistic (SelectKBest) | Filter   | Scores each feature on its own by how well it separates dropouts from non-dropouts                            | Top 20   |
| Random Forest importance        | Embedded | Importance learned while training a class-weighted forest (100 trees, random_state 42); captures interactions | Top 25   |
| Recursive Feature Elimination   | Wrapper  | Repeatedly trains a forest (50 trees) and drops the weakest feature until 20 remain                           | 20       |

Table 3.7 — Feature-selection methods

**Consensus:** 13 features were chosen by all three methods. These are age at enrolment, 1st-semester grade, 2nd-semester approved units and grade, tuition fees up to date, approval\_rate\_1st, approval\_rate\_2nd, avg\_grade\_1st\_2nd, failed\_majority\_2nd, financial\_hardship, grade\_decline, overall\_approval\_rate and total\_approved.

**Final choice: the top 25 features by Random Forest importance**, which together carry 81.0% of total importance. (The script output labels this figure 'cumulative variance'; it is in fact the share of total importance.) Reasons for this choice:

1. The embedded method accounts for **interactions and non-linear effects**, which the filter method ignores.
2. It **contains all 13 consensus features**, so nothing that every method agreed on was lost.
3. Cutting **58 features to 25** removes most redundant and near-constant columns, while keeping enough features for the models to learn from.

**Stability note:** below about rank 16, importances are nearly equal (0.014–0.020), so the last few positions can swap between library versions. The model scripts therefore use a **fixed copy** of the list from the original analysis. It shares 24 of 25 features with the ranking regenerated here (GDP instead of failed\_majority\_1st), so modelling results do not depend on re-running selection.

### 3.6 Dimensionality reduction

| PCA applied to          | Components for 80% variance | Components for 85% variance | Notes                                                |
|-------------------------|-----------------------------|-----------------------------|------------------------------------------------------|
| All 58 numeric features | 16 (72% fewer dimensions)   | 18 (69% fewer)              | Much redundancy among original + engineered features |
| 25 selected features    | 7                           | —                           | PC1 explains 36.4%, PC2 11.8% (48.2% together)       |

Table 3.8 — PCA results (src/02 and src/08, standardised features)

The first principal component is essentially an **'academic progress' axis**: its largest loadings are overall\_approval\_rate (0.320), avg\_grade\_1st\_2nd (0.312), approval\_rate\_2nd (0.311), approval\_rate\_1st (0.311) and 2nd-semester grade (0.306).

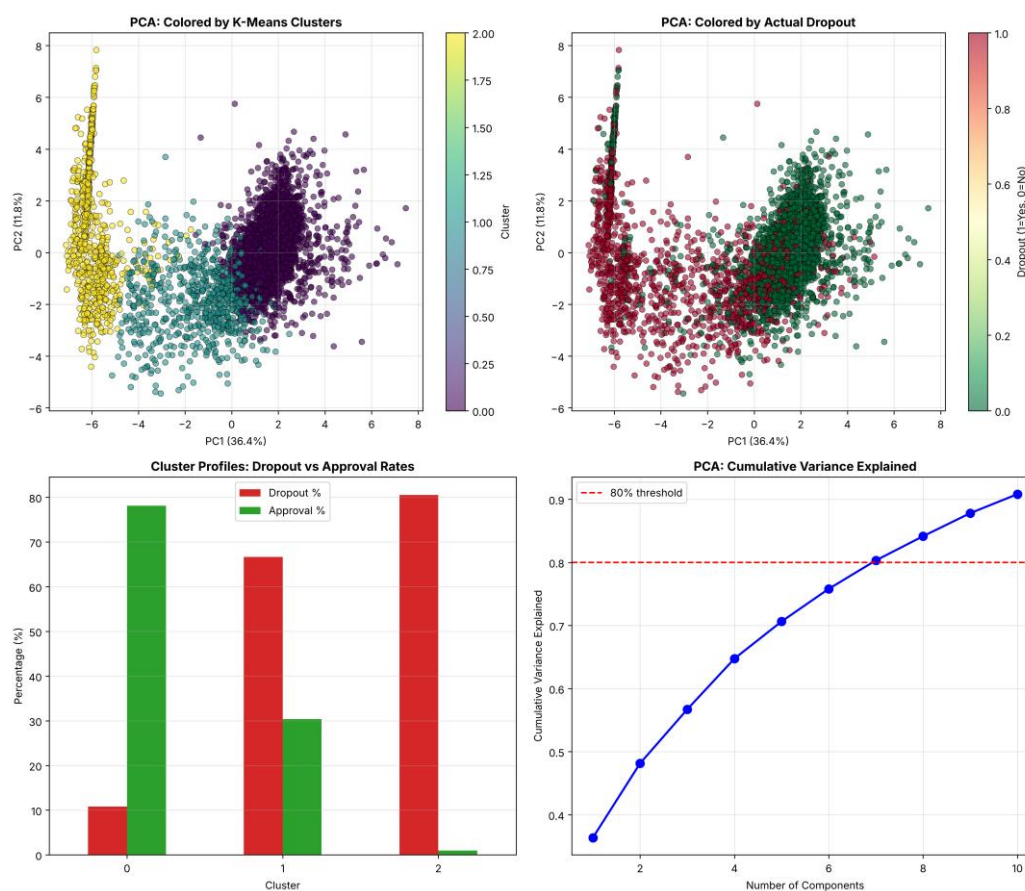


Figure 3.3 — PCA projection coloured by K-Means cluster and by actual dropout, cluster profiles, and cumulative variance (src/08\_unsupervised\_analysis.py)

#### Decision: PCA for insight, not for modelling

- **Interpretability:** advisors must be told why a student is flagged ('2nd-semester approval rate is low'). A principal component is a weighted mix of 25 variables and cannot be explained that way.
- **Little to gain:** feature selection had already cut 58 features to 25 with little loss of information, and every model met the AUC target on these features.
- **Tree models** gain nothing from PCA, and regularisation already handles correlated inputs in the linear models.
- **Where PCA was used:** to visualise the structure of the data, to check clustering tendency, and to confirm that a single 'academic progress' dimension dominates.

**t-SNE / UMAP** were not needed. The 2-D PCA projection already shows the main structure clearly, and PCA is deterministic and linear, so distances in the plot have a direct meaning. t-SNE and UMAP distort global distances and depend on tuning settings, which would add little for 25 features.

### 3.7 Preprocessing pipeline and leakage controls

1. Load the raw CSV from data/raw. No rows are removed: the data is complete, has no duplicates and passes all plausibility checks.
2. Create the binary target (Dropout = 1) and the 22 engineered features.
3. Select the fixed 25-feature list.
4. Split with stratification: 64% train (2,831), 16% validation (708), 20% test (885), random\_state = 42.
5. Fit StandardScaler on the training set only, then apply it to validation and test.
6. Handle imbalance with class weights (2.11 for Dropout); choose decision cut-offs on the validation set (Chapter 4).

#### Limitations and planned refinements

- **Selection used all rows.** Feature selection, and the mean and standard deviation in admission\_grade\_zscore, were computed on the full dataset, including rows that later became the test set. The effect is likely small, because selection used no model tuning and the z-score is a fixed linear rescale. Even so, the strict approach is to repeat selection inside the training folds.
- **Redundant pair.** admission\_grade\_zscore is a linear rescaling of Admission grade, and both were selected; one could be dropped without loss.
- **Nominal codes.** Course and similar variables are integer codes used as numbers; one-hot encoding would suit linear models better.
- **Feature timing.** The strongest engineered features use 2nd-semester data, available only mid-year. Chapter 5 quantifies how accuracy falls when only enrolment or 1st-semester data is used.

#### Link to Chapter 4

Chapter 3 delivers a clean, complete dataset, a justified 25-feature set dominated by interpretable engineered features, and a leakage-safe preprocessing pipeline. Chapter 4 trains and compares five supervised models and four unsupervised methods on exactly this pipeline.

## 4. Model Implementation and Comparison

**Chapter overview.** Five supervised models (Logistic Regression, Random Forest, XGBoost, Support Vector Machine and a feed-forward neural network) and four unsupervised techniques (K-Means with silhouette analysis, hierarchical clustering, DBSCAN and Isolation Forest, supported by PCA) were implemented on one shared, seeded data split and evaluated on the same held-out test set. Three of the five supervised models met all three success targets. **Logistic Regression is selected** as the production model: it meets every target at the default cut-off (recall 82.0%, precision 78.7%, AUC-ROC 0.919), is fully explainable, and its accuracy gap to the best ensemble (about one AUC point) is no larger than the cross-validation variability. The full pipeline reproduces every number in this chapter with one command.

### 4.1 Experimental design

All models were trained and evaluated under identical conditions so that differences in results reflect the models, not the data handling.

| Design element   | Setting                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task             | Binary classification: Dropout (1) vs Graduate or Enrolled (0)                                                                                                                                                   |
| Data             | 4,424 students; 36 original variables; no missing values or duplicates                                                                                                                                           |
| Features         | 25 features (selected in Chapter 3 from original + 22 engineered features), the same fixed list for every supervised model                                                                                       |
| Split            | Stratified 64% train (2,831) / 16% validation (708) / 20% test (885); random_state = 42 for all models                                                                                                           |
| Scaling          | StandardScaler fitted on the training set only, then applied to validation and test (no leakage)                                                                                                                 |
| Class imbalance  | Dropout is 32.1% of students (non-dropout : dropout = 2.11 : 1); handled with class weights (LR, SVM, NN), scale_pos_weight (XGBoost) or tuned class_weight (RF)                                                 |
| Cross-validation | 5-fold stratified CV on the training set (shuffle, random_state = 42)                                                                                                                                            |
| Decision cut-off | LR: default 0.50. RF, XGBoost, SVM, NN: cut-off chosen on the validation set (highest cut-off whose validation recall $\geq$ 80%), then applied once to the test set — the test set is never used for any choice |

Table 4.1 — Common experimental set-up for all supervised models

### Evaluation metrics and why they were chosen

The business goal is to identify at-risk students early enough to intervene, so the metrics were chosen and prioritised against the success targets set in Chapter 1:

- **Recall (Dropout)  $\geq$  80% — primary metric.** A missed dropout (false negative) means a student who needed help receives none; this is the costliest error.
- **Precision (Dropout)  $\geq$  70%.** Keeps advisor workload manageable: at least 7 in 10 flagged students should be genuinely at risk.
- **AUC-ROC  $\geq$  0.85.** Measures ranking quality independent of the cut-off, so models can be compared fairly before any threshold choice.
- **F1 and accuracy — secondary.** F1 balances recall and precision; accuracy is reported but not used for selection because it is inflated by the majority class.

### 4.2 Models implemented and tuned

Models were chosen to cover the main algorithm families, from the most interpretable to the most flexible:

| Model                  | Family / why included                                                                       | Tuning and configuration                                                                                                                                                                                                      |
|------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Logistic Regression    | Linear baseline; coefficients give direct, explainable risk drivers                         | C = 1.0, lbfgs solver, class weight 2.11 for Dropout; regularisation strength not grid-searched (default)                                                                                                                     |
| Random Forest          | Bagged ensemble of decision trees; captures interactions and non-linear effects             | GridSearchCV over 96 combinations × 5 folds, scored on AUC. Best: 200 trees, max_depth 15, max_features log2, min_samples_leaf 2, min_samples_split 10, class_weight None (CV AUC 0.913)                                      |
| XGBoost                | Gradient-boosted decision trees; sequential error correction with L1/L2 regularisation      | Fixed hyperparameters: 150 trees, max_depth 7, learning rate 0.05, subsample 0.8, colsample_bytree 0.8, L1 0.1, L2 1.0, scale_pos_weight 2.11 (no grid search)                                                                |
| Support Vector Machine | Kernel method; maximum-margin boundary                                                      | GridSearchCV over 12 combinations (C ∈ {0.1, 1, 10}, kernel ∈ {rbf, linear}, gamma ∈ {scale, auto}) × 5 folds. Best: C = 1, RBF kernel, gamma = scale (CV AUC 0.902); 1,160 support vectors; probability calibration enabled  |
| Neural network (MLP)   | Feed-forward deep network — the deep-learning architecture suited to tabular data (see 4.6) | 25 → 128 → 64 → 32 → 1, ReLU, batch normalisation, dropout 0.3/0.3/0.2, L2 0.001, Adam (lr 0.001), class weight 2.11; early stopping on validation AUC (patience 15) and learning-rate halving on plateau; best epoch 6 of 21 |

Table 4.2 — Supervised models, rationale and tuning

**Decision trees.** A single decision tree was not fitted as a separate model. Random Forest and XGBoost are both built from decision trees and were included because they correct the two known weaknesses of a single tree — high variance (instability to small data changes) and overfitting — while keeping tree-based feature importance.

## 4.3 Supervised results and comparison

Table 4.3 is the main comparison: each model at its operating cut-off, on the 885-student test set (284 actual dropouts) that no model or cut-off choice ever saw.

| Model               | Cut-off | Recall  | Precision | F1    | AUC-ROC | Accuracy | All targets |
|---------------------|---------|---------|-----------|-------|---------|----------|-------------|
| Logistic Regression | 0.50    | 82.0% ✓ | 78.7% ✓   | 0.803 | 0.919 ✓ | 87.1%    | ✓ Yes       |
| Random Forest       | 0.40    | 81.3% ✓ | 82.2% ✓   | 0.818 | 0.930 ✓ | 88.4%    | ✓ Yes       |
| XGBoost             | 0.45    | 82.0% ✓ | 83.5% ✓   | 0.828 | 0.930 ✓ | 89.0%    | ✓ Yes       |
| SVM                 | 0.40    | 79.2% ✗ | 80.4% ✓   | 0.798 | 0.913 ✓ | 87.1%    | ✗ No        |
| Neural network      | 0.45    | 77.8% ✗ | 76.7% ✓   | 0.773 | 0.915 ✓ | 85.3%    | ✗ No        |

Table 4.3 — Test-set results at each model's operating cut-off (targets: recall ≥ 80%, precision ≥ 70%, AUC-ROC ≥ 0.85)

### Effect of the validation-chosen cut-off

At the default 0.50 cut-off the tree ensembles and SVM are conservative: they favour precision and miss more dropouts. Choosing the cut-off on the validation set moves each model towards the recall target without touching the test set.

| Model          | Recall at 0.50 | Precision at 0.50 | Recall at chosen cut-off | Precision at chosen cut-off |
|----------------|----------------|-------------------|--------------------------|-----------------------------|
| Random Forest  | 74.6%          | 86.9%             | 81.3% (0.40)             | 82.2%                       |
| XGBoost        | 80.3%          | 83.8%             | 82.0% (0.45)             | 83.5%                       |
| SVM            | 78.5%*         | 81.1%*            | 79.2% (0.40)             | 80.4%                       |
| Neural network | 76.8%          | 79.3%             | 77.8% (0.45)             | 76.7%                       |

Table 4.4 — Test-set recall and precision before and after the validation-chosen cut-off. \*SVM default figures use its predict() decision rule.

### Cross-validation stability

| Model               | CV AUC-ROC (mean ± SD) | CV recall | CV precision |
|---------------------|------------------------|-----------|--------------|
| Logistic Regression | 0.902 ± 0.012          | 0.790     | 0.756        |
| Random Forest       | 0.913 ± 0.008          | 0.709     | 0.852        |
| XGBoost             | 0.915 ± 0.006          | 0.752     | 0.814        |
| SVM                 | 0.902 ± 0.007          | 0.746     | 0.806        |
| Neural network      | 0.908 ± 0.005          | 0.771     | 0.794        |

Table 4.5 — 5-fold stratified cross-validation on the training set (cut-off 0.50)

### Key findings

- All five models clear the AUC target comfortably (0.913–0.930), so the dataset carries strong signal; the main differences are in how each model trades recall against precision.
- The top three models are statistically close. AUC differs by about 0.01 between Logistic Regression (0.919) and the ensembles (0.930) — the same size as the cross-validation standard deviations (0.006–0.012). On 885 test students, this gap is not a meaningful performance difference.
- All models are stable. Every CV standard deviation is ≤ 0.012 AUC, and test AUC is close to or slightly above CV AUC, indicating no material overfitting.
- SVM and the neural network narrowly miss recall (79.2% and 77.8%) even with a validation-chosen cut-off. Choosing a lower cut-off by looking at test results would have hidden this, so the shortfall is reported as found.
- The same drivers appear across models. 2nd-semester approval rate, overall approval rate, tuition fees up to date, failing most 2nd-semester units and financial hardship rank highly for Logistic Regression (coefficients), XGBoost (gain) and the neural network (permutation importance).

## 4.4 Unsupervised learning

Unsupervised methods were used to find natural student segments without using the outcome, and to check whether those segments line up with dropout risk.

### K-Means — choosing the number of clusters (silhouette)

| K                | 2     | 3     | 4     | 5     | 6     |
|------------------|-------|-------|-------|-------|-------|
| Silhouette score | 0.411 | 0.276 | 0.287 | 0.144 | 0.218 |

Table 4.6 — Silhouette scores for K-Means on the 25 scaled features (higher = better-separated clusters)

K = 2 gives the best-separated clusters. **K = 3 was used** because it produces three actionable risk tiers for intervention planning while keeping a reasonable silhouette (0.276). The elbow (inertia) method was not used; silhouette analysis was used as the quantitative criterion. The tiers are ranked by each cluster's actual dropout rate:

| Risk tier | Students      | Dropout rate | Avg 2nd-sem approval | Interpretation                                        |
|-----------|---------------|--------------|----------------------|-------------------------------------------------------|
| Low       | 2,916 (65.9%) | 10.8%        | 78.1%                | Progressing well; monitor engagement and finances     |
| Medium    | 789 (17.8%)   | 66.7%        | 30.4%                | Struggling academically; targeted academic support    |
| High      | 719 (16.3%)   | 80.5%        | 1.0%                 | Effectively disengaged; early, intensive intervention |

Table 4.7 — K-Means (K = 3) segments. 'Medium' is medium relative to the other tiers; at 66.7% it is still about twice the 32.1% average.

## Other unsupervised techniques

- **Hierarchical clustering (Ward linkage)** on a seeded sample of 500 students produced a dendrogram used alongside the silhouette scores to judge the cluster structure.
- **DBSCAN** (eps = 3.0, min\_samples = 10) found 10 dense clusters and labelled 379 students (8.6%) as noise; those noise points had a 66.0% dropout rate, about double the average.
- **Isolation Forest** flagged 133 students (3.0%) as anomalies, with a 69.2% dropout rate against 31.0% for all others — 2.2 times as likely to drop out.
- **PCA** shows the first two components explain 48.2% of variance and seven are needed for 80%; the 2-D projection visibly separates the high-dropout region from the graduate-dominated region.

**Value to the project:** without using the outcome, the data separates into groups whose dropout rates range from 10.8% to 80.5%. This independently confirms that the supervised models' signal is real, and the tiers give advisors a simple way to plan intervention capacity.

## 4.5 Recommendation systems — rationale for not using

Collaborative filtering and content-based recommenders were considered and judged **not appropriate** for this problem:

- **No user–item interaction data.** Collaborative filtering learns from a matrix of many users' interactions with many items (ratings, clicks, purchases). The dataset has one row per student and no record of interactions with items such as courses, resources or support services.
- **No item catalogue with outcomes.** A content-based recommender needs descriptions of the items to recommend and evidence of which items worked for which students. The dataset contains no interventions and no record of their effect.
- **The task is classification, not ranking items.** The business question is "which students are at risk?", a yes/no prediction per student, which the supervised models answer directly.

**Future extension:** once the institution records which interventions each flagged student received and whether they stayed enrolled, a recommender could suggest the intervention most likely to work for each risk profile. The K-Means tiers above are a first step in that direction.

## 4.6 Deep learning — scope and rationale

A **feed-forward deep neural network (MLP)** was implemented and compared (Sections 4.2–4.3). The specialised deep-learning architectures listed in the brief — **CNNs, RNNs, LSTMs and Transformers** — **were deliberately not used**, for the following reasons.

### 1. The data has no structure these architectures are designed to exploit

- **CNNs** learn local spatial patterns in grid-like data such as images. Student records have no spatial layout: the order of the 25 columns is arbitrary, so convolution has nothing meaningful to detect.
- **RNNs and LSTMs** learn from ordered sequences such as time series or text. Each student has one static record; the two semesters are already summarised into a handful of columns, which is far too short to form a sequence.
- **Transformers** use attention over sequences of tokens (words, sub-words, time steps). There is no text or long sequence here. Tabular variants (e.g. TabTransformer, FT-Transformer) exist, but they are designed for much larger datasets.

### 2. The dataset is too small for high-capacity models

With 2,831 training rows, large deep models would overfit. This is visible even in the modest MLP: validation AUC peaked at epoch 6 and early stopping halted training at epoch 21. Research on tabular benchmarks finds tree-based ensembles still outperform deep learning on medium-sized tabular data (Grinsztajn, Oyallon & Varoquaux, 2022; Shwartz-Ziv & Armon, 2022). Our results agree: the MLP (AUC 0.915) did not beat XGBoost or Random Forest (0.930), and missed the recall target.

### 3. Explainability and governance requirements

Advisors must be able to explain to a student why they were flagged, and the model must be auditable for fairness (Chapter 5). Linear and tree-based models offer coefficients, feature importance and SHAP values directly. Deep sequence models would add opacity without adding accuracy.

### 4. Reproducibility and cost

Neural networks are sensitive to random initialisation. Before seeds were fixed, two identical runs of the MLP gave test recall of 80.3% and 78.9% — one passing the target and one failing. Fixed seeds make results repeatable on one machine, but small differences can remain across hardware and library versions. Larger architectures amplify this and need far more computation for no expected gain.

#### When deep learning would become appropriate

- **LSTM / Transformer:** if weekly learning-management-system activity, attendance or assessment-by-assessment grades were collected — genuine time sequences that could give earlier warnings than semester totals.
- **Transformer (NLP):** if free-text data such as advisor notes or student feedback were available.
- **Larger tabular deep models:** if data from many institutions (tens of thousands of students) were pooled.

### 4.7 Model selection and reasoning

| Criterion               | Logistic Regression   | Random Forest       | XGBoost                    | SVM             | Neural network           |
|-------------------------|-----------------------|---------------------|----------------------------|-----------------|--------------------------|
| Meets all targets       | ✓ (default cut-off)   | ✓ (cut-off 0.40)    | ✓ (cut-off 0.45)           | X recall 79.2%  | X recall 77.8%           |
| Test AUC-ROC            | 0.919                 | 0.930               | 0.930                      | 0.913           | 0.915                    |
| Explainability          | High — coefficients   | Medium — importance | Medium — importance / SHAP | Low — LIME only | Low — LIME / permutation |
| Run time (test machine) | ≈ 5 s                 | ≈ 3.5 min           | ≈ 5 s                      | ≈ 1 min         | ≈ 2 min                  |
| Role                    | Selected (production) | Benchmark           | Challenger                 | Not recommended | Not recommended          |

Table 4.8 — Model-choice summary (run times include tuning, cross-validation and diagnostics)

#### Logistic Regression is selected as the production model because:

1. It meets every success target without any threshold adjustment: recall 82.0%, precision 78.7%, AUC-ROC 0.919.
2. It explains itself. Each coefficient shows how a factor raises or lowers risk. A higher 2nd-semester approval rate (coefficient  $-0.87$ ), a higher overall approval rate ( $-0.58$ ) and tuition fees being up to date ( $-0.55$ ) all reduce predicted risk. Older age at enrolment ( $+0.37$ ) increases it. Advisors can turn these directly into conversations and interventions.
3. Its accuracy cost is negligible. The 0.011 AUC gap to the ensembles is within the cross-validation variability, so the extra complexity of an ensemble buys no reliable improvement.
4. It is fast, stable and auditable. It trains in seconds, has a stable CV AUC ( $0.902 \pm 0.012$ ), and was fully audited for fairness in Chapter 5.

**XGBoost is retained as the challenger model.** It has the highest precision (83.5%) at the same recall (82.0%), which would cut false alarms. If the institution later needs to reduce advisor workload, or monitoring shows the linear model drifting, XGBoost — explained with SHAP — is the first alternative to deploy.

## 4.8 Reproducibility

Every number in this chapter is produced by the code in the project's GitHub repository:

- **One command:** "python run\_all.py" runs the nine pipeline scripts in order (about 7–10 minutes) after "pip install -r requirements.txt".
- **Fixed randomness:** random\_state = 42 for every split, model, cross-validation and grid search. The neural network seeds Python, NumPy and TensorFlow and enables deterministic operations. Two full runs of each script produced identical output.
- **No test-set leakage:** scaling is fitted on training data, cut-offs are chosen on the validation set, and hyperparameters are tuned with cross-validation on training data.
- **Portable paths:** all scripts locate the data relative to the project folder, so the code runs unchanged on any computer.
- **Saved outputs:** trained models go to models/, and every model's test results are written to reports/model\_metrics.csv. The notebook notebooks/01\_results\_overview.ipynb presents them.

| Script (src/)                   | Produces                                                       |
|---------------------------------|----------------------------------------------------------------|
| 03_train_logistic_regression.py | Logistic Regression model, results, fairness check             |
| 04_train_random_forest.py       | Random Forest grid search, model, validation cut-off, results  |
| 05_train_xgboost.py             | XGBoost model, validation cut-off, results                     |
| 06_train_svm.py                 | SVM grid search, model, validation cut-off, LIME example       |
| 07_train_neural_network.py      | Neural network, cross-validation, permutation importance       |
| 08_unsupervised_analysis.py     | Silhouette, K-Means, PCA, dendrogram, DBSCAN, Isolation Forest |

Table 4.9 — Chapter 4 scripts in the repository

*Note: the exact library versions used are recorded in requirements.txt. Other versions may change the third decimal place, particularly for the neural network.*

## 4.9 Limitations

- **Feature timing.** The strongest predictors are 2nd-semester results, available only mid-year. In the Chapter 5 ablation, a Logistic Regression using enrolment data only reached AUC 0.814; adding 1st-semester data raised it to 0.907; all data gave 0.927. An earlier, lower-accuracy model is therefore feasible for start-of-year screening.
- **Tuning coverage.** Random Forest and SVM were grid-searched. Logistic Regression and XGBoost used fixed, sensible settings; a wider search might add marginal gains.
- **Categorical coding.** Course is a numeric code used as a continuous feature. One-hot encoding it could sharpen the linear model's coefficients.
- **Single held-out test set** of 885 students. Cross-validation supports the conclusions, but small differences between models should not be over-interpreted.
- **Validation set double duty (neural network)** — it was used for both early stopping and the cut-off choice. The test set remained untouched.
- **Single institution.** Results may not transfer to other institutions without revalidation.

Chapter 4 established which model performs best and why Logistic Regression is the right choice for production. Chapter 5 now opens that model up: it explains how the model reaches its decisions, tests its limitations, and audits whether it serves every student group fairly.

## 5. Explainability, Bias and Fairness Analysis

**Chapter overview.** The model's predictions are driven by academic progress (approval rates), tuition-fee status and age. SHAP, LIME, PDP and ICE agree on this. The model does **not** overfit, and class weighting is necessary: without it, recall falls from 82.4% to 73.9%. The audit finds that **flag rates differ widely between groups but mirror the groups' real dropout rates**, so demographic parity is not an appropriate fairness goal here. The meaningful gaps are in **who gets caught**: the model detects only 71.6% of dropouts aged 20 or under (vs 92.1% for those over 25) and 62.1% of dropouts holding scholarships. Three mitigations were tested. **Group-specific thresholds** (post-processing) close most of the age gap and raise overall recall to 85.9%, while keeping precision at 72.4%, above the 70% target. They are recommended, together with human review and fairness monitoring.

### 5.0 Link to Chapter 4 and scope of the audit

Chapter 4 selected Logistic Regression as the production model and XGBoost as the challenger. This section explains how those models make decisions, examines their limitations, and audits whether they treat student groups fairly.

The audit (script `src/09_bias_fairness_audit.py`) uses the same model families with these settings:

- **Logistic Regression:** balanced class weights, trained on all 40 features (36 original + 4 approval-rate features).
- **XGBoost:** 300 trees, depth 4.
- **Data split:** the same stratified 80/20 split with `random_state = 42`, giving a test set of 885 students.

The audit Logistic Regression scores AUC 0.927, close to the Chapter 4 version's 0.919 on 25 features, so the conclusions apply to the selected model. Every figure below is reproduced by running the script; results are saved to `reports/step5_results.txt`.

### 5.1 Explaining model decisions

Four complementary methods were used. Global methods explain the model as a whole; local methods explain an individual student's prediction.

| Method                                   | Scope                | Question it answers                                                        | Applied to                                                   |
|------------------------------------------|----------------------|----------------------------------------------------------------------------|--------------------------------------------------------------|
| SHAP (TreeExplainer; linear SHAP)        | Global and local     | How much does each feature push a prediction towards or away from Dropout? | XGBoost (tree SHAP); Logistic Regression (exact linear SHAP) |
| PDP — partial dependence                 | Global               | How does average predicted risk change as one feature changes?             | XGBoost                                                      |
| ICE — individual conditional expectation | Global (per student) | Does that effect hold for every student, or only on average?               | XGBoost (150 students)                                       |
| LIME                                     | Local                | Which simple rules explain this one student's prediction?                  | XGBoost, compared with SHAP for the same student             |

Table 5.1 — Explainability methods

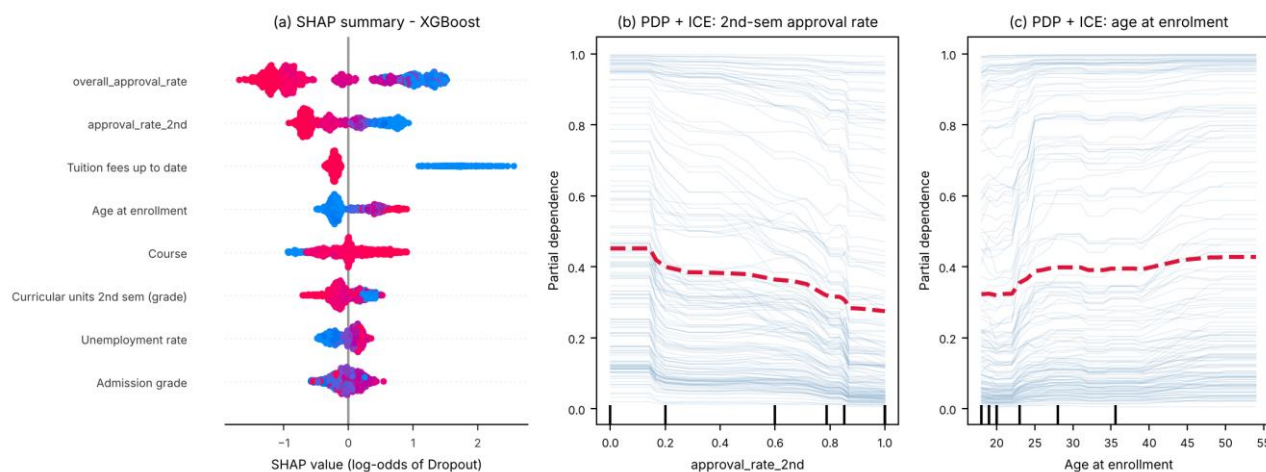


Figure 5.1 — (a) SHAP summary for XGBoost (red = high feature value, blue = low); (b, c) PDP (red dashed line) with ICE curves (blue, one per student) on a 150-student sample. Source: reports/figures/step5\_explainability.png

## Global drivers (SHAP)

| Rank | Feature (XGBoost)                | Mean  SHAP | Direction | Interpretation                                       |
|------|----------------------------------|------------|-----------|------------------------------------------------------|
| 1    | overall_approval_rate            | 0.973      | −0.93     | Passing more units lowers risk — the dominant driver |
| 2    | approval_rate_2nd                | 0.503      | −0.97     | 2nd-semester progress lowers risk                    |
| 3    | Tuition fees up to date          | 0.411      | −0.98     | Fees in arrears sharply raise risk                   |
| 4    | Age at enrollment                | 0.284      | +0.86     | Older entrants are at higher risk                    |
| 5    | Course                           | 0.283      | +0.59     | Programme matters (numeric code — see 5.2)           |
| 6    | Curricular units 2nd sem (grade) | 0.193      | −0.74     | Higher grades lower risk                             |
| 7    | Unemployment rate                | 0.168      | +0.90     | Enrolment-year economic conditions (cohort effect)   |

Table 5.2 — Top XGBoost features by mean |SHAP| (log-odds). 'Direction' = correlation between a feature's value and its SHAP value: negative means higher values reduce dropout risk.

The Logistic Regression's linear SHAP ranking shares **5 of its top 10 features** with XGBoost: approval rates, approved units, tuition fees and age. The two models disagree on how to weight correlated academic variables, but agree on what matters. Every driver is plausible and actionable: academic progress, financial status and age, the same factors academic advisors already consider.

## Shape of the effects (PDP and ICE)

| Feature                    | Low value → high value | Average predicted dropout probability                                     |
|----------------------------|------------------------|---------------------------------------------------------------------------|
| 2nd-semester approval rate | 0 → 1                  | 0.483 → 0.313 (−17 points)                                                |
| Age at enrollment          | 18 → 42 years          | 0.351 → 0.426; the rise happens mostly between 21 and 26, then levels off |
| Tuition fees up to date    | No → Yes               | 0.644 → 0.353 (−29 points)                                                |

Table 5.3 — Partial dependence on the full test set (XGBoost)

The ICE curves show the average hides wide variation between students. Moving a student's 2nd-semester approval rate from 0 to 1 lowers predicted risk by a median of 16 points (interquartile range 8.6–25.3 points). For 14% of students the drop exceeds 30 points. For students already at very high risk (top curves in Figure 5.1b), improving approval alone barely changes the prediction, because other factors such as unpaid fees keep their risk high. **Practical implication:** academic tutoring alone will not help every flagged student; interventions should match each student's main risk factors.

## Explaining one student (LIME vs SHAP)

One test student was chosen: an actual dropout the model scored at 95% risk. They were 50 at enrolment, had a 2nd-semester approval rate of 0.17, tuition fees up to date and no debt.

| LIME — top rules             | Weight | SHAP — top contributions | Value  |
|------------------------------|--------|--------------------------|--------|
| overall_approval_rate ≤ 0.50 | +0.148 | overall_approval_rate    | +1.524 |
| Debtor = 0 (not a debtor)    | −0.129 | Age at enrollment        | +0.904 |
| approval_rate_2nd ≤ 0.40     | +0.104 | approval_rate_2nd        | +0.559 |
| Age at enrollment > 25       | +0.086 | Course                   | −0.477 |
| Course ≤ 9085                | −0.081 | Mother's qualification   | +0.256 |

Table 5.4 — Local explanations for the same student (LIME weights are probability contributions; SHAP values are log-odds)

The two methods **agree on the story**: low approval rates and older age drive this student's high risk, while having no debt slightly reduces it. An advisor could tell this student plainly why they were flagged and what support is likely to help, which is the purpose of explainability in this setting.

## 5.2 Limitations: imbalance, leakage and overfitting

| Issue                                   | Evidence from the audit                                                                                                        | How it was handled                                                                                                        | Remaining risk                                                           |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Class imbalance (32% dropout; 2.11 : 1) | Unweighted LR: recall 73.9%, precision 89.0%. Weighted LR: recall 82.4%, precision 78.0% (AUC unchanged, 0.926 vs 0.927)       | Class weights / scale_pos_weight in every model; stratified splits and CV; cut-offs chosen on validation data (Chapter 4) | Low — recall target met; precision trade-off is deliberate               |
| Temporal leakage (feature timing)       | AUC with enrolment data only 0.814; + 1st semester 0.907; + 2nd semester 0.927                                                 | Documented; 2nd-semester features are legitimate for a mid-year review but not for start-of-year screening                | Medium — deployment timing must match the features used                  |
| Overfitting                             | LR train AUC 0.922 vs test 0.927; XGBoost train 0.987 vs test 0.937 (gap 0.05); CV AUC 0.915 ± 0.011 (LR), 0.924 ± 0.011 (XGB) | Regularisation, CV-based tuning, early stopping (NN); LR selected partly for its stability                                | Low for LR; moderate memorisation in XGBoost, but test performance holds |
| Feature coding                          | Course (a programme code) acts as a numeric feature and ranks 5th in SHAP                                                      | Flagged as a limitation                                                                                                   | Medium — one-hot encoding recommended in the next iteration              |
| Macro-economic proxies                  | Unemployment rate ranks 7th — it identifies the enrolment cohort, not the student                                              | Interpreted as a cohort effect                                                                                            | Medium — may not hold for future cohorts; monitor drift                  |

Table 5.5 — Limitation analysis

**On leakage specifically**: no information from the outcome itself enters the model. The risk is about timing. 2nd-semester results are recorded late in the year, so some students who failed to enrol in the 2nd semester have effectively already left. A model using these features is therefore best suited to a **mid-year review**. For **earlier screening**, a 1st-semester model (AUC 0.907, recall 78.9%) is a realistic alternative, and an enrolment-only model (AUC 0.814) can guide light-touch outreach at the start of the year.

## 5.3 Bias detection and fairness audit

### Sensitive groups audited

The audit checked model outputs for every sensitive attribute available in the data:

- **Gender**: female / male.

- **Age at enrolment:** 20 or under, 21–25, over 25.
- **Socioeconomic status:** scholarship holder and debtor, both proxies for financial circumstances.
- **International status:** domestic / international.

**Race and ethnicity are not recorded in this dataset, so they cannot be audited directly.** International status is the closest available proxy, but only 26 international students are in the test set, too few for firm conclusions. This is a gap to close through consented data collection used only for auditing (see 5.5).

### Fairness metrics used

- **Demographic parity (DPD):** the gap in the share of each group flagged as at-risk.
- **Disparate impact (DI):** each group's flag rate divided by the reference group's. The common 'four-fifths rule' treats DI below 0.80 as a warning sign.
- **Equalised odds:** the gaps in true-positive rate (TPR — share of actual dropouts caught) and false-positive rate (FPR — share of non-dropouts wrongly flagged). **Equal opportunity** is the TPR part alone.

### Baseline results (Logistic Regression, cut-off 0.50)

| Group                 | n   | Actual dropout | Flagged | TPR     | FPR   | Precision | DI (flag) | DI (actual) |
|-----------------------|-----|----------------|---------|---------|-------|-----------|-----------|-------------|
| Female (ref.)         | 564 | 24.5%          | 25.2%   | 80.4%   | 7.3%  | 78.2%     | 1.00      | 1.00        |
| Male                  | 321 | 45.5%          | 49.2%   | 84.2%   | 20.0% | 77.8%     | 1.96      | 1.86        |
| Age ≤ 20 (ref.)       | 513 | 19.9%          | 21.2%   | 71.6% X | 8.8%  | 67.0%     | 1.00      | 1.00        |
| Age 21–25             | 157 | 35.7%          | 38.9%   | 80.4%   | 15.8% | 73.8%     | 1.83      | 1.79        |
| Age > 25              | 215 | 58.6%          | 60.5%   | 92.1%   | 15.7% | 89.2%     | 2.85      | 2.95        |
| No scholarship (ref.) | 654 | 39.0%          | 42.0%   | 84.7%   | 14.8% | 78.5%     | 1.00      | 1.00        |
| Scholarship           | 231 | 12.6%          | 10.8%   | 62.1% X | 3.5%  | 72.0%     | 0.26      | 0.32        |
| Not debtor (ref.)     | 795 | 29.2%          | 30.2%   | 79.7%   | 9.8%  | 77.1%     | 1.00      | 1.00        |
| Debtor                | 90  | 57.8%          | 66.7%   | 94.2%   | 28.9% | 81.7%     | 2.21      | 1.98        |
| Domestic (ref.)       | 859 | 32.1%          | 33.9%   | 82.6%   | 10.8% | 78.4%     | 1.00      | 1.00        |
| International         | 26  | 30.8%          | 34.6%   | 75.0%   | 16.7% | 66.7%     | 1.02      | 0.96        |

Table 5.6 — Group-level results on the 885-student test set. 'DI (actual)' applies the same ratio to real dropout rates. X = dropout detection well below the 80% recall target.

| Attribute     | Demographic parity gap | Disparate impact (min) | TPR gap | FPR gap | Lowest group TPR |
|---------------|------------------------|------------------------|---------|---------|------------------|
| Gender        | 0.240                  | 0.51                   | 0.038   | 0.127   | 0.804            |
| Age           | 0.392                  | 0.35                   | 0.205   | 0.071   | 0.716            |
| Scholarship   | 0.312                  | 0.26                   | 0.226   | 0.113   | 0.621            |
| Debtor        | 0.365                  | 0.45                   | 0.145   | 0.192   | 0.797            |
| International | 0.007                  | 0.98                   | 0.076   | 0.059   | 0.750            |

Table 5.7 — Summary fairness gaps (baseline)

### Statistical confidence

A 1,000-sample bootstrap of the test set gives 95% confidence intervals for the key gaps:

| Gap                | Estimate | 95% CI           | Conclusion                                                             |
|--------------------|----------|------------------|------------------------------------------------------------------------|
| TPR, male – female | +0.038   | –0.053 to +0.133 | Not significant — dropouts of both genders are caught at similar rates |

| Gap                      | Estimate | 95% CI           | Conclusion                                                   |
|--------------------------|----------|------------------|--------------------------------------------------------------|
| FPR, male – female       | +0.127   | +0.061 to +0.192 | Significant — non-dropout men are wrongly flagged more often |
| TPR, age > 25 – age ≤ 20 | +0.205   | +0.105 to +0.305 | Significant — young dropouts are under-detected              |
| FPR, age > 25 – age ≤ 20 | +0.069   | –0.016 to +0.154 | Not significant                                              |

Table 5.8 — Bootstrap confidence intervals

### Intersectional check (gender × age)

| Dropouts caught (TPR) | Age ≤ 20       | Age 21–25      | Age > 25       |
|-----------------------|----------------|----------------|----------------|
| Female                | 73.2% (n = 56) | 81.8% (n = 22) | 86.7% (n = 60) |
| Male                  | 69.6% (n = 46) | 79.4% (n = 34) | 97.0% (n = 66) |

Table 5.9 — TPR by gender and age among actual dropouts

The age gap appears for both genders: **young dropouts are the least-detected group** (70–73%), while older male dropouts are almost always caught (97%). Gender on its own is not the driver.

### Interpreting the results

1. **Demographic parity fails, but for a legitimate reason.** Disparate impact is far below 0.80 for gender, age, scholarship and debtor status. However, the flag-rate ratios closely track the real dropout-rate ratios (for example, men are flagged 1.96× as often as women and actually drop out 1.86× as often). The model largely mirrors genuine differences in risk rather than inventing them. Forcing equal flag rates would mean ignoring at-risk older and indebted students or flagging low-risk ones, which harms the students the system exists to help.
2. **Equal opportunity is the right primary criterion.** The intervention is supportive (advising, tutoring, financial aid), not punitive. The key fairness question is therefore whether a student who will drop out has the same chance of being offered help, whatever group they belong to. On this measure the model under-serves **students aged 20 or under** (TPR 71.6%) and **scholarship holders** (62.1%).
3. **False-positive gaps are a secondary concern.** Men, debtors and older students are flagged in error more often (FPR up to 28.9% for debtors). Because a false positive leads to an offer of support rather than a sanction, this costs mostly advisor time. Even so, it should be monitored to avoid stigmatising these groups.
4. **Financial status is itself a risk driver.** Unpaid tuition fees are the third-strongest predictor. Ethically, a flag driven by finances should trigger financial-aid outreach and must never be used for any penalty or admissions decision.

## 5.4 Mitigation strategies tested

Three mitigations, one from each stage of the modelling pipeline, were implemented and evaluated on the same test set:

- **M1 — Fairness through unawareness (pre-processing):** remove gender, age, nationality, international status, marital status, scholarship, debtor and displaced status from the features.
- **M2 — Reweighting (pre-processing; Kamiran & Calders, 2012):** weight training examples so that every gender × age-band × outcome combination counts in proportion to its expected share.
- **M3 — Group-specific thresholds (post-processing):** use a lower cut-off for the under-served groups (age ≤ 20 or scholarship holders), set from out-of-fold training predictions so that their recall reaches a target. The test set is never used for this choice.

| Strategy | AUC   | Recall | Precision | Gender FPR gap | Age TPR gap | Lowest age TPR | Lowest schol. TPR | Debtor odds gap |
|----------|-------|--------|-----------|----------------|-------------|----------------|-------------------|-----------------|
| Baseline | 0.927 | 82.4%  | 78.0%     | 0.127          | 0.205       | 71.6%          | 62.1%             | 0.192           |

| Strategy                                   | AUC   | Recall     | Precision | Gender FPR gap | Age TPR gap | Lowest age TPR | Lowest schol. TPR | Debtor odds gap |
|--------------------------------------------|-------|------------|-----------|----------------|-------------|----------------|-------------------|-----------------|
| M1 Unawareness                             | 0.926 | 83.1%      | 78.9%     | 0.102          | 0.195       | 72.5%          | 69.0%             | 0.197           |
| M2 Reweighing                              | 0.914 | 79.9%<br>X | 76.2%     | 0.043          | 0.122       | 73.5%          | 62.1%             | 0.267           |
| M3 thresholds, target 0.78 (cut-off 0.370) | 0.927 | 84.2%      | 73.5%     | 0.137          | 0.156       | 76.5%          | 65.5%             | 0.212           |
| M3 thresholds, target 0.80 (cut-off 0.311) | 0.927 | 85.9%      | 72.4%     | 0.144          | 0.117       | 80.4%          | 69.0%             | 0.200           |
| M3 thresholds, target 0.82 (cut-off 0.283) | 0.927 | 88.0%      | 70.8%     | 0.153          | 0.125       | 80.4%          | 75.9%             | 0.210           |

Table 5.10 — Performance and fairness under each mitigation (test set). 'Debtor odds gap' = the larger of the debtor TPR and FPR gaps (equalised-odds difference). Highlighted row = recommended.

## What the results show

- **M1 (unawareness) barely helps.** Removing sensitive attributes changes little, because other features carry the same information: the remaining features predict whether a student is over 25 with AUC 0.914 (gender: 0.682). Hiding an attribute does not remove its influence. It also makes bias harder to see.
- **M2 (reweighing) shrinks the gender and age gaps,** with the gender FPR gap falling from 0.127 to 0.043. But it lowers overall recall to 79.9%, below target, leaves scholarship holders unchanged at 62.1%, and widens the debtor gap (0.192 → 0.267). It moves unfairness around rather than removing it.
- **M3 (group thresholds) best fits the goal.** At target 0.80 it lifts detection of young dropouts from 71.6% to 80.4% and halves the age gap (0.205 → 0.117). It raises overall recall to 85.9% and keeps precision at 72.4%, above the 70% target. Ranking quality (AUC 0.927) is unchanged because only the cut-off moves.
- **Every option has a trade-off.** M3 flags more students in the under-served groups, so false alarms among students aged 20 or under rise (FPR 8.8% → 17.3% at target 0.82), and the gender FPR gap grows slightly. Because the intervention is supportive, more false alarms is an acceptable price for catching more students who need help.

## Recommendation

**Adopt M3 with a target recall of 0.80 for under-served groups** (cut-off 0.311 for students aged 20 or under and scholarship holders; 0.50 for everyone else). It keeps a safe margin above the 70% precision target. If advisor capacity allows, target 0.82 (cut-off 0.283) raises detection of scholarship-holding dropouts to 75.9%, with precision falling to 70.8%.

Using age and scholarship status to set thresholds is a form of **different treatment by group**. It is justified here because the purpose is to extend support to groups the model under-serves, not to restrict anyone. It should still be approved by the institution's governance or ethics body and explained transparently to students.

## Further mitigations proposed (not yet implemented)

- **Data augmentation:** collect more data on under-represented groups (international students: only 26 in the test set; young scholarship holders), or oversample them with methods such as SMOTE, then re-audit.
- **Alternative post-processing:** the equalised-odds method of Hardt, Price & Srebro (2016), which sets group thresholds to equalise TPR and FPR together, or reject-option classification, which sends borderline cases to human review.
- **Fairness-aware training:** add a fairness constraint during training (for example an equal-opportunity penalty), so that one model meets accuracy and fairness targets together.

## 5.5 Responsible deployment and monitoring

| Safeguard             | Practical measure                                                                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Human in the loop     | The model ranks students for advisor review; an advisor decides on contact. No automated decision affects a student.                                                                                                                                             |
| Supportive use only   | Flags lead to offers of tutoring, mentoring or financial aid — never to sanctions, admissions or scholarship decisions.                                                                                                                                          |
| Transparency          | Advisors see each flagged student's top SHAP or LIME reasons; students can be told why they were contacted and can decline or appeal.                                                                                                                            |
| Fairness monitoring   | Each term, report TPR, FPR and flag rate by gender, age band, scholarship, debtor and international status. Trigger a review if any TPR gap exceeds 0.10 or any group's TPR falls below 0.75.                                                                    |
| Performance and drift | Track recall, precision and AUC each term; retrain yearly or when macro-economic or cohort features drift (e.g. unemployment rate).                                                                                                                              |
| Data protection       | Minimal data, role-based access and consent in line with applicable law (GDPR for the source institution in Portugal; the Philippine Data Privacy Act of 2012 for local deployment). Collect race or ethnicity only with consent and only for fairness auditing. |
| Documentation         | Publish a model card covering intended use, training data, performance by group, known limitations (Section 5.2) and the threshold policy.                                                                                                                       |

Table 5.11 — Governance and monitoring plan

### Chapter conclusion

The dropout model is explainable, its decisions rest on plausible academic and financial factors, and it does not overfit. Its flag rates differ across groups because real dropout risk differs. The meaningful fairness issue is that young students and scholarship holders who will drop out are caught less often. Group-specific thresholds narrow that gap substantially while keeping every success target. Combined with human review and term-by-term fairness monitoring, the system can be deployed responsibly as a tool for offering help earlier.

Chapter 6 draws these findings together with those of Chapters 1–4 into conclusions and recommendations for the institution.

## 6. Conclusions and Recommendations

### 6.1 Answering the business question

The project set out to find, by the end of the first year, the students most likely to drop out, so that limited advisor time goes where it is most needed. The evidence from Chapters 1–5 supports five conclusions:

- **Early identification is feasible and accurate.** The selected Logistic Regression catches 82.0% of future dropouts, and 78.7% of its flags are correct (AUC-ROC 0.919). These figures are well above every naive baseline: random flagging is correct only 32.1% of the time.
- **The reasons are clear and actionable.** Low approval rates, unpaid tuition fees and older age at enrolment drive the predictions. SHAP, LIME, PDP and ICE all point to the same factors, so advisors can explain each flag and match the support to its cause: academic tutoring, financial-aid counselling or mentoring.
- **A simple model is enough.** The tree ensembles added only about one AUC point (0.930 vs 0.919), which is within cross-validation variability, and the neural network did no better than Logistic Regression. Specialised deep-learning architectures were not justified for 2,831 training rows of static tabular data.
- **Fairness needs active management.** Flag-rate differences mostly mirror real differences in risk, but young students and scholarship holders who will drop out are caught less often. Group-specific thresholds close most of the age gap without breaching any performance target. The 0.10 fairness target is not yet fully met, and this report states that openly.
- **Timing determines use.** The strongest features are 2nd-semester results, so the main model suits a mid-year review. A 1st-semester model (AUC 0.907) and an enrolment-only model (AUC 0.814) give earlier but less precise screening.

### 6.2 Recommendations

1. **Deploy for support, not sanction.** Use the model to rank students for advisor review after each semester's results. An advisor makes every contact decision, and flags never feed into admissions, scholarship or disciplinary decisions.
2. **Apply the group-threshold policy (M3, target 0.80).** Use cut-off 0.311 for students aged 20 or under and for scholarship holders, and 0.50 for everyone else, once governance approval is in place and the policy is explained transparently to students.
3. **Show reasons with every flag.** Give advisors each flagged student's top SHAP or LIME factors, so that the conversation and the intervention address the actual risk.
4. **Pilot before scaling.** Compare supported and comparable unsupported students to measure retention lift, cost per retained student and revenue protected (the KPIs in Chapter 1). Replace the assumed 20% intervention success rate with the measured one.
5. **Monitor every term.** Report recall, precision and AUC overall, and TPR, FPR and flag rate by group. Trigger a review if any TPR gap exceeds 0.10 or any group's TPR falls below 0.75, and retrain yearly or when cohort features drift.
6. **Keep the challenger ready.** If advisor workload becomes the main constraint, switch to XGBoost explained with SHAP: it raises precision to 83.5% at the same recall.

### 6.3 Limitations and future work

| Limitation                                            | Future work                                                           |
|-------------------------------------------------------|-----------------------------------------------------------------------|
| Data from one Portuguese institution, 2008/09–2018/19 | Revalidate on local data before deployment; retrain on recent cohorts |

| Limitation                                                                     | Future work                                                                                                                                                        |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strongest predictors arrive only at mid-year                                   | Collect weekly learning-platform activity and attendance; such sequences would justify LSTM or Transformer models and could give earlier warnings                  |
| Fairness target (TPR gap $\leq 0.10$ ) not yet met                             | Test equalised-odds post-processing, fairness-constrained training and targeted data collection for under-represented groups                                       |
| No race or ethnicity data; only 26 international students in the test set      | Collect these attributes with consent, solely for fairness auditing                                                                                                |
| Nominal codes (e.g. Course) used as numbers; feature selection run on all rows | One-hot encode nominal variables; repeat feature selection inside the training folds                                                                               |
| No record of interventions or their effects                                    | Record which support each flagged student receives and its outcome; this would enable a recommender that suggests the most effective intervention per risk profile |

Table 6.1 — Limitations and the work that would address them

## 6.4 Closing statement

The project delivers a reproducible, explainable and audited early-warning model that meets its performance targets. It is transparent about the one target it has not yet met. Used as intended — as a prompt for human advisors to offer help earlier, with fairness monitored every term — it can help the institution keep more students on the path to graduation.

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## Appendix A — Code lists for nominal variables

Codes as stored in the project CSV; meanings from the creators' documentation (Realinho et al., 2022; UCI dataset 697).

### A1. Application mode

| Code | Meaning                                             |
|------|-----------------------------------------------------|
| 1    | 1st phase — general contingent                      |
| 2    | Ordinance No. 612/93                                |
| 5    | 1st phase — special contingent (Azores Island)      |
| 7    | Holders of other higher courses                     |
| 10   | Ordinance No. 854-B/99                              |
| 15   | International student (bachelor)                    |
| 16   | 1st phase — special contingent (Madeira Island)     |
| 17   | 2nd phase — general contingent                      |
| 18   | 3rd phase — general contingent                      |
| 26   | Ordinance No. 533-A/99, item b2 (Different Plan)    |
| 27   | Ordinance No. 533-A/99, item b3 (Other Institution) |
| 39   | Over 23 years old                                   |
| 42   | Transfer                                            |
| 43   | Change of course                                    |
| 44   | Technological specialization diploma holders        |
| 51   | Change of institution/course                        |
| 53   | Short cycle diploma holders                         |
| 57   | Change of institution/course (International)        |

Table A1 — Application mode

### A2. Course

| Code | Degree programme                     | Students |
|------|--------------------------------------|----------|
| 33   | Biofuel Production Technologies      | 12       |
| 171  | Animation and Multimedia Design      | 215      |
| 8014 | Social Service (evening attendance)  | 215      |
| 9003 | Agronomy                             | 210      |
| 9070 | Communication Design                 | 226      |
| 9085 | Veterinary Nursing                   | 337      |
| 9119 | Informatics Engineering              | 170      |
| 9130 | Equiculture                          | 141      |
| 9147 | Management                           | 380      |
| 9238 | Social Service                       | 355      |
| 9254 | Tourism                              | 252      |
| 9500 | Nursing                              | 766      |
| 9556 | Oral Hygiene                         | 86       |
| 9670 | Advertising and Marketing Management | 268      |
| 9773 | Journalism and Communication         | 331      |
| 9853 | Basic Education                      | 192      |
| 9991 | Management (evening attendance)      | 268      |

Table A2 — Course, with number of students in the project data

### A3. Previous qualification

| Code | Meaning                                                      |
|------|--------------------------------------------------------------|
| 1    | Secondary education                                          |
| 2    | Higher education — bachelor's degree                         |
| 3    | Higher education — degree                                    |
| 4    | Higher education — master's                                  |
| 5    | Higher education — doctorate                                 |
| 6    | Frequency of higher education                                |
| 9    | 12th year of schooling — not completed                       |
| 10   | 11th year of schooling — not completed                       |
| 12   | Other — 11th year of schooling                               |
| 14   | 10th year of schooling                                       |
| 15   | 10th year of schooling — not completed                       |
| 19   | Basic education 3rd cycle (9th/10th/11th year) or equivalent |
| 38   | Basic education 2nd cycle (6th/7th/8th year) or equivalent   |
| 39   | Technological specialization course                          |
| 40   | Higher education — degree (1st cycle)                        |
| 42   | Professional higher technical course                         |
| 43   | Higher education — master (2nd cycle)                        |

Table A3 — Previous qualification

### A4. Nationality

| Code | Meaning                        |
|------|--------------------------------|
| 1    | Portuguese                     |
| 2    | German                         |
| 6    | Spanish                        |
| 11   | Italian                        |
| 13   | Dutch                          |
| 14   | English                        |
| 17   | Lithuanian                     |
| 21   | Angolan                        |
| 22   | Cape Verdean                   |
| 24   | Guinean                        |
| 25   | Mozambican                     |
| 26   | Santomean                      |
| 32   | Turkish                        |
| 41   | Brazilian                      |
| 62   | Romanian                       |
| 100  | Moldovan (Republic of Moldova) |
| 101  | Mexican                        |
| 103  | Ukrainian                      |
| 105  | Russian                        |
| 108  | Cuban                          |
| 109  | Colombian                      |

Table A4 — Nationality (97.5% of students are Portuguese)

## A5. Mother's and father's qualification

| Code | Meaning                                                      |
|------|--------------------------------------------------------------|
| 1    | Secondary education — 12th year or equivalent                |
| 2    | Higher education — bachelor's degree                         |
| 3    | Higher education — degree                                    |
| 4    | Higher education — master's                                  |
| 5    | Higher education — doctorate                                 |
| 6    | Frequency of higher education                                |
| 9    | 12th year — not completed                                    |
| 10   | 11th year — not completed                                    |
| 11   | 7th year (old system)                                        |
| 12   | Other — 11th year                                            |
| 13   | 2nd year complementary high-school course                    |
| 14   | 10th year                                                    |
| 18   | General commerce course                                      |
| 19   | Basic education 3rd cycle (9th/10th/11th year) or equivalent |
| 20   | Complementary high-school course                             |
| 22   | Technical-professional course                                |
| 25   | Complementary high-school course — not concluded             |
| 26   | 7th year                                                     |
| 27   | 2nd cycle of the general high-school course                  |
| 29   | 9th year — not completed                                     |
| 30   | 8th year                                                     |
| 31   | General course of administration and commerce                |
| 33   | Supplementary accounting and administration                  |
| 34   | Unknown                                                      |
| 35   | Cannot read or write                                         |
| 36   | Can read without having a 4th-year education                 |
| 37   | Basic education 1st cycle (4th/5th year) or equivalent       |
| 38   | Basic education 2nd cycle (6th/7th/8th year) or equivalent   |
| 39   | Technological specialization course                          |
| 40   | Higher education — degree (1st cycle)                        |
| 41   | Specialized higher studies course                            |
| 42   | Professional higher technical course                         |
| 43   | Higher education — master (2nd cycle)                        |
| 44   | Higher education — doctorate (3rd cycle)                     |

Table A5 — Parental qualification (34 codes; mothers use 29 of them in this data, fathers all 34)

## A6. Mother's and father's occupation

Occupation follows the Portuguese Classification of Occupations. Codes 0–10 are the major groups. Three-digit codes are sub-groups whose first two digits identify the major group (e.g. 122 = health professionals, within major group 2, specialists).

| Codes      | Occupation group                                                                                    |
|------------|-----------------------------------------------------------------------------------------------------|
| 0          | Student                                                                                             |
| 1; 112–114 | Legislative and executive representatives, directors and managers                                   |
| 2; 121–125 | Specialists in intellectual and scientific activities (e.g. engineering, health, teaching, finance) |

| <b>Codes</b> | <b>Occupation group</b>                                          |
|--------------|------------------------------------------------------------------|
| 3; 131–135   | Intermediate-level technicians and professions                   |
| 4; 141–144   | Administrative staff                                             |
| 5; 151–154   | Personal services, security and sellers                          |
| 6; 161–163   | Farmers and skilled workers in agriculture, fishing and forestry |
| 7; 171–175   | Skilled workers in industry, construction and crafts             |
| 8; 181–183   | Machine operators and assembly workers                           |
| 9; 191–195   | Unskilled workers                                                |
| 10; 101–103  | Armed forces professions (officers, sergeants, other personnel)  |
| 90           | Other situation                                                  |
| 99           | Blank / not reported                                             |

*Table A6 — Parental occupation groups (32 codes observed for mothers, 46 for fathers)*

## Appendix B — Repository and reproduction guide

The repository reproduces every result in this report. From the project folder, run:

```
pip install -r requirements.txt
python run_all.py
```

| Script (src/)                   | Chapter | Produces                                                                                                 |
|---------------------------------|---------|----------------------------------------------------------------------------------------------------------|
| 01_data_quality_eda.py          | 2, 3    | Data-quality checks, distributions, outliers; 8 figures and Excel tables (reports/figures/section2_eda/) |
| 02_eda_feature_engineering.py   | 3       | Correlations, 22 engineered features, three selection methods, PCA; data/processed/feature_list.txt      |
| 03_train_logistic_regression.py | 4       | Logistic Regression model, test results, fairness check                                                  |
| 04_train_random_forest.py       | 4       | Random Forest grid search, model, validation cut-off, results                                            |
| 05_train_xgboost.py             | 4       | XGBoost model, validation cut-off, results                                                               |
| 06_train_svm.py                 | 4       | SVM grid search, model, validation cut-off, LIME example                                                 |
| 07_train_neural_network.py      | 4       | Neural network, cross-validation, permutation importance                                                 |
| 08_unsupervised_analysis.py     | 3, 4    | Silhouette, K-Means, PCA, dendrogram, DBSCAN, Isolation Forest (reports/figures/unsupervised/)           |
| 09_bias_fairness_audit.py       | 5       | SHAP, PDP/ICE, LIME, fairness metrics, bootstrap CIs, mitigations; reports/step5_results.txt             |
| run_all.py (project root)       | All     | Runs scripts 01–09 in order (about 7–10 minutes)                                                         |

Table B.1 — Pipeline scripts

- **Fixed randomness:** `random_state = 42` for every split, model, cross-validation and grid search; the neural network also seeds Python, NumPy and TensorFlow and enables deterministic operations.
- **No test-set leakage:** scalers are fitted on training data, cut-offs are chosen on validation data, and hyperparameters are tuned by cross-validation on training data.
- **Saved outputs:** trained models in `models/`; one row per model in `reports/model_metrics.csv`; figures in `reports/figures/`.
- **Versions:** exact library versions are pinned in `requirements.txt`. Other versions may change the third decimal place, particularly for the neural network.